From Zero to the ϕ^4 Laboratory

A self-contained primer on the mathematics behind structured denoising

Kalman filtering and smoothing · Fourier and Hermite analysis · EM and its generalisations · Fisher information and quasi-likelihood · hidden Markov models, Markov random fields and belief propagation · the Allen–Cahn equation and its stochastic version · and how they meet

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Every figure is generated from the project's own code (`tools/make_primer_figs.py`); every number quoted from an experiment names the experiment and says whether it is frozen or a preview.

How to read this primer

The project asks one question: given many noisy copies of sequences that come from a one-dimensional physical system, can we learn the system's local rule and then compute the *exact* best denoiser? Answering it uses ten pieces of mathematics that are usually taught in ten different courses. This primer builds each of them from the ground up, in an order where every chapter only uses what came before.

- **Part I, foundations** (§1–3). The calculus, linear algebra and probability you need, including the one trick used everywhere (completing the square), conditional expectation as the best predictor, and Markov chains.
- **Part II, inference on chains** (§4–6). Noise, scores and Tweedie's formula; the Kalman filter and smoother; hidden Markov models, Markov random fields and belief propagation.
- **Part III, spectral tools** (§7–8). Fourier analysis (why noise erases fine detail) and Hermite polynomials (why fourth-order structure fades like e^{-4t}).
- **Part IV, learning** (§9–10). Maximum likelihood, the EM algorithm and its general forms; Fisher information, the Cramér–Rao bound, quasi-likelihood.
- **Part V, physics** (§11–12). From a ball rolling in a valley to the Allen–Cahn equation and its stochastic version; the transfer operator that turns a field into a Markov chain.
- **Part VI, the new theory** (§13). The tilted walk, reversibility, the learned pair energy, and what the experiments have measured.

Three kinds of boxes recur. *Key idea* boxes state the one thing to remember. *In our project* boxes say where the idea lives in the code and experiments. *Careful* boxes flag the traps that have cost us time. Proofs are given whenever they are short enough to be read in a minute; the longer ones are sketched with the step that matters.

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1 The toolbox

1.1 Slopes, curvature, and the second difference

A function f assigns a number $f(x)$ to every number x. Its *derivative* $f'(x)$ is the slope of the graph at x: for a small step h,

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}.$$

The second derivative $f''(x)$ is the slope of the slope, which measures curvature: positive where the graph bends upwards like a cup, negative where it bends down like a cap. Taylor's formula says that near x every smooth function looks like a parabola,

$$f(x+h) = f(x) + h\,f'(x) + \tfrac{1}{2}h^2 f''(x) + \tfrac{1}{6}h^3 f'''(x) + \dots \tag{1}$$

Write (1) once with $+h$ and once with $-h$ and add. The odd powers cancel, and

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + O(h^2). \tag{2}$$

The left side is the *second difference*. Rewrite it as $\frac{2}{h^2}\big[\frac{1}{2}(f(x+h) + f(x-h)) - f(x)\big]$: it is the gap between the average of the two neighbours and the value at the point. It is positive when the point sits below its neighbours, negative when it sits above.

Key idea

The second difference (2) is all a *Laplacian* is in one dimension. When physicists write $\partial_x^2\phi$ or $\nabla^2\phi$ for a quantity ϕ that varies along a line, they mean "how far ϕ at a point is below the average of its neighbours", in the limit of close neighbours. In more dimensions one adds the same quantity for each direction. We will meet it as the force of the springs that hold a string together (§11).

A function of several variables $f(x_1, \dots, x_n)$ has *partial derivatives* $\partial f/\partial x_i$: the slope in direction i with the others held fixed. Collected in a vector they form the *gradient* ∇f, which points in the direction in which f increases fastest.

1.2 Integrals as sums, and the grid

The integral $\int_a^b f(x)\,\mathrm{d}x$ is the area under the graph. Chop $[a, b]$ into $N_g - 1$ intervals of width δ with points $u_1, \dots, u_{N_g}$; then

$$\int_a^b f(x)\,\mathrm{d}x \approx \sum_{k=1}^{N_g} w_k\, f(u_k), \qquad w_k = \delta \text{ except } w_1 = w_{N_g} = \delta/2. \tag{3}$$

This is the *trapezoid rule*; its error shrinks like δ^2. Every integral in this project is done this way: on $N_g = 401$ points spread over $[-8, 8]$, so $\delta = 0.04$.

1.3 Exponentials, logarithms, and why we compute in logs

The exponential turns sums into products, $e^{a+b} = e^a e^b$, and the logarithm turns products back into sums. Probabilities of long sequences are products of many numbers smaller than one, which underflow a computer (in double precision e^{-800} is exactly zero). So we store logarithms, and to add numbers stored as logs we use the *log-sum-exp* identity

$$\log \sum_k e^{y_k} = m + \log \sum_k e^{y_k - m}, \qquad m = \max_k y_k,$$

which never overflows because every exponent on the right is at most zero.

1.4 Vectors, matrices, eigenvectors

A vector is a list of numbers, a matrix a table. The product of two matrices is

$$(AB)_{kj} = \sum_m A_{km} B_{mj}, \tag{4}$$

a sum over an intermediate index m. Remember this shape: "sum over the state in the middle" is the one operation that belief propagation repeats (§6). The transpose $A^\top$ swaps rows and columns; A is *symmetric* if $A = A^\top$. The inverse A^{-1} undoes A.

An *eigenvector* v of a square matrix A is a direction that A only stretches: $Av = \lambda v$, with *eigenvalue* λ. Two facts carry most of the weight in this primer.

- A symmetric matrix has real eigenvalues $\lambda_0 \geq \lambda_1 \geq \dots$ and eigenvectors $v_0, v_1, \dots$ that are mutually perpendicular and of unit length. It can be rebuilt from them, $A = \sum_n \lambda_n v_n v_n^\top$, and so $A^N = \sum_n \lambda_n^N v_n v_n^\top$. For large N the largest $|\lambda_n|$ dominates: *repeated multiplication forgets everything except the leading eigenvector*. That is the power method, and it is the reason a long chain "forgets" its start.
- *Perron–Frobenius*: if every entry of A is positive, the largest eigenvalue λ_0 is positive and unrepeated, and its eigenvector can be chosen with every entry positive.

A symmetric A is *positive definite* if $v^\top A v > 0$ for every nonzero v, equivalently if all its eigenvalues are positive. Covariance matrices are of this kind.

1.5 The Gaussian integral and completing the square

The single most useful integral is

$$\int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}. \tag{5}$$

(Square it, read the square as an integral over the plane, and switch to polar coordinates; the angular integral gives 2π and the radial one gives 1.) The single most useful trick is *completing the square*:

$$-\tfrac{A}{2}x^2 + Bx = -\tfrac{A}{2}\Big(x - \tfrac{B}{A}\Big)^2 + \tfrac{B^2}{2A} \qquad (A > 0). \tag{6}$$

It says that any exponent that is quadratic in x describes a bell curve centred at B/A with width $1/\sqrt{A}$. Products of Gaussian bumps are Gaussian bumps because their exponents add, and the sum of two quadratics is a quadratic. The whole of the Kalman filter is this sentence applied repeatedly.

Check yourself.

1.1 Show that the second difference of $f(x) = x^3$ equals $6x$ exactly, for every h. Why is there no error term?

1.2 The matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has positive entries. Find its eigenvalues and eigenvectors, and say in which direction $A^N v$ points for large N, whatever the starting vector v (unless v is special).

2 Probability from scratch

2.1 Random variables, densities, expectations

A random quantity a is described by its *density* $p(a)$: the probability that a lands in a tiny interval $[u, u + \mathrm{d}u]$ is $p(u)\,\mathrm{d}u$, and $\int p = 1$. The *expectation* of any function of a is the weighted average

$$\mathbb{E}[f(a)] = \int f(u)\,p(u)\,\mathrm{d}u.$$

The mean is $\mu = \mathbb{E}[a]$, the variance $\mathrm{Var}(a) = \mathbb{E}[(a-\mu)^2]$, the standard deviation its square root. For two quantities the covariance $\mathrm{Cov}(a,b) = \mathbb{E}[(a - \mathbb{E}a)(b - \mathbb{E}b)]$ measures whether they move together, and the correlation divides it by both standard deviations, landing in $[-1, 1]$.

Two random quantities have a *joint* density $p(a,b)$. Integrating one away gives the *marginal*, $p(a) = \int p(a,b)\,\mathrm{d}b$. Fixing one gives the *conditional*,

$$p(a \mid b) = \frac{p(a,b)}{p(b)},$$

the law of a once b is known. They are independent when $p(a,b) = p(a)p(b)$. Writing the joint both ways, $p(a \mid x)p(x) = p(x \mid a)p(a)$, gives *Bayes' rule*:

$$\underbrace{p(a \mid x)}_{\text{posterior}} = \frac{\overbrace{p(x \mid a)}^{\text{likelihood}}\ \overbrace{p(a)}^{\text{prior}}}{\underbrace{p(x)}_{\text{evidence}}} \ \propto\ p(x \mid a)\,p(a). \tag{7}$$

The symbol $\propto$ ("proportional to") means "equal up to a factor that does not depend on a"; the factor is recovered at the end by requiring the density to integrate to one.

2.2 The Gaussian, in one and many dimensions

The Gaussian (normal) density with mean m and variance s^2 is $\mathcal{N}(u; m, s^2) = \exp\big(-(u-m)^2/2s^2\big)/\sqrt{2\pi s^2}$; by (5) it integrates to one. A weighted sum of independent Gaussians is Gaussian, with means and variances combining as $\mathbb{E}[\alpha a + \beta z] = \alpha\mathbb{E}a + \beta\mathbb{E}z$ and $\operatorname{Var}(\alpha a + \beta z) = \alpha^2 \operatorname{Var} a + \beta^2 \operatorname{Var} z$.

A vector $a = (a_1, \dots, a_n)$ is Gaussian with mean vector m and covariance matrix Σ ($\Sigma_{ij} = \operatorname{Cov}(a_i, a_j)$) if

$$p(a) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp\Big(-\tfrac{1}{2}(a-m)^\top \Sigma^{-1}(a-m)\Big). \tag{8}$$

The inverse Σ^{-1} is the *precision matrix*, and its zeros have a meaning that will matter: if $(\Sigma^{-1})_{ij} = 0$, then a_i and a_j are independent *once all the other coordinates are known*.

Example 2.1 (the Gaussian AR(1) chain has a tridiagonal precision)**.** Let $a_1 \sim \mathcal{N}(0,1)$ and $a_{i+1} = \rho a_i + \varepsilon_i$ with independent $\varepsilon_i \sim \mathcal{N}(0, 1-\rho^2)$. The joint density is $p(a_1)\prod_i \mathcal{N}(a_{i+1}; \rho a_i, 1-\rho^2)$, whose exponent is $-\frac{1}{2}a_1^2 - \sum_i (a_{i+1} - \rho a_i)^2/2(1-\rho^2)$. Each term couples only neighbours, so the precision matrix has nonzeros only on the diagonal and the two adjacent diagonals, while the covariance $\Sigma_{ij} = \rho^{|i-j|}$ is full. *Local in the precision, global in the covariance*: this is the whole reason inference on chains is cheap.

2.3 Conditioning a Gaussian on a noisy observation

Suppose $a \sim \mathcal{N}(m, P)$ and we observe $x = \alpha a + \sqrt{\Delta}\, z$ with independent noise $z \sim \mathcal{N}(0,1)$. By Bayes' rule the posterior is proportional to the prior times the likelihood:

$$p(a \mid x) \propto \exp\Big(-\frac{(a-m)^2}{2P} - \frac{(x-\alpha a)^2}{2\Delta}\Big).$$

The exponent is quadratic in a. Collect the coefficient of a^2 (that is $-\frac{1}{2}A$ with $A = 1/P + \alpha^2/\Delta$) and of a (that is $B = m/P + \alpha x/\Delta$) and complete the square (6):

$$\boxed{\frac{1}{P'} = \frac{1}{P} + \frac{\alpha^2}{\Delta}, \qquad m' = P'\Big(\frac{m}{P} + \frac{\alpha x}{\Delta}\Big)} \tag{9}$$

The posterior is Gaussian. *Precisions add* (prior precision plus the information in the measurement), and the posterior mean is a precision-weighted average of what we believed and what we saw. The same result in "gain" form, which is how the Kalman filter is usually written, is

$$m' = m + k\,(x - \alpha m), \qquad P' = (1 - k\alpha)P, \qquad k = \frac{\alpha P}{\alpha^2 P + \Delta}. \tag{10}$$

The *innovation* $x - \alpha m$ is the surprise in the measurement; the gain k says how much of it to believe. (Check: $(1-k\alpha)P = P\Delta/(\alpha^2 P + \Delta)$, which is the reciprocal of $1/P + \alpha^2/\Delta$.)

For vectors the statement is the same with matrices. If (a, x) is jointly Gaussian with blocks $\Sigma_{aa}, \Sigma_{ax}, \Sigma_{xx}$, then

$$\mathbb{E}[a \mid x] = m_a + \Sigma_{ax}\Sigma_{xx}^{-1}(x - m_x), \qquad \operatorname{Cov}(a \mid x) = \Sigma_{aa} - \Sigma_{ax}\Sigma_{xx}^{-1}\Sigma_{xa}. \tag{11}$$

Notice that the conditional mean is a *linear* function of x. This is special to Gaussians.

2.4 The best predictor and the best linear predictor

Suppose we must guess a from x with a function $g(x)$ and pay the squared error. The best g is the *conditional mean* $g^\star(x) = \mathbb{E}[a \mid x]$. Proof: for any g,

$$\mathbb{E}\big[(a-g)^2\big] = \mathbb{E}\big[(a-g^\star)^2\big] + \mathbb{E}\big[(g^\star - g)^2\big],$$

because the cross term $\mathbb{E}[(a-g^\star)(g^\star - g)]$ vanishes (given x, the second factor is fixed and the first has mean zero). The second term is never negative, so nothing beats $g^\star$. This is the *minimum mean-square error* (MMSE) estimator, and it is what the word "denoiser" means in this primer.

If we restrict ourselves to *linear* guesses $g(x) = Gx$, the best one is $G = \operatorname{Cov}(a, x)\operatorname{Var}(x)^{-1}$, the LMMSE (linear MMSE) estimator. It depends only on covariances. For a Gaussian pair it coincides with $\mathbb{E}[a \mid x]$ by (11); for anything else, the true conditional mean is nonlinear and strictly better.

Key idea

Every law has a *covariance twin*: the Gaussian with the same means and covariances. The twin's denoiser is the LMMSE estimator of the true law. The gap between the twin's denoiser and the true one is exactly what the non-Gaussian structure of the data is worth to a denoiser. Most of this project measures that gap.

2.5 Moments, cumulants, kurtosis

For a standardised quantity (mean 0, variance 1) the third moment $\mathbb{E}a^3$ is the *skewness* (asymmetry) and $\mathbb{E}a^4$ the *kurtosis*, which is 3 for a Gaussian. The *excess kurtosis* $\mathbb{E}a^4 - 3$ is positive for heavy tails (a Laplace law has $+3$) and negative for laws with "shoulders" or two humps (the ϕ^4 marginal at $\Theta = 0.3$ has -1.35).

These are special cases of *cumulants* κ_n, defined as the coefficients of $\log \mathbb{E}e^{sa} = \sum_n \kappa_n s^n/n!$. The first is the mean, the second the variance, and for a standardised law κ_3 and κ_4 are skewness and excess kurtosis. Two properties make cumulants the right language. For independent a and b they add, $\kappa_n(a+b) = \kappa_n(a) + \kappa_n(b)$, and they scale as $\kappa_n(ca) = c^n \kappa_n(a)$. A Gaussian has $\kappa_n = 0$ for every $n \geq 3$; this is what "non-Gaussian" will mean.

A first look at the decay that organises the whole project: if $x = \alpha a + \sqrt{\Delta} z$ with Gaussian z independent of a, then for $n \geq 3$

$$\kappa_n(x) = \alpha^n \kappa_n(a) + \Delta^{n/2} \kappa_n(z) = \alpha^n \kappa_n(a). \tag{12}$$

With $\alpha = e^{-t}$ (§4), the n-th cumulant fades like e^{-nt}: skewness like e^{-3t}, kurtosis like e^{-4t}, while the variance is held fixed. Chapter 8 turns this into a statement about whole functions.

Check yourself.

2.1 The prior is $a \sim \mathcal{N}(0,1)$ and we observe $x = a + z$ with $z \sim \mathcal{N}(0,1)$ ($\alpha = \Delta = 1$). What is the posterior?

2.2 A standardised law has excess kurtosis -1.35. What is the excess kurtosis of $x = e^{-t}a + \sqrt{1-e^{-2t}}\, z$ at $t = 1$?

3 Markov chains

3.1 Definition

A sequence $a_1, a_2, \ldots, a_L$ is a *Markov chain* if the next value depends on the past only through the present: $p(a_{i+1} \mid a_1, \ldots, a_i) = p(a_{i+1} \mid a_i)$. The joint density then factorises,

$$p(a_1, \ldots, a_L) = \mu(a_1) \prod_{i=1}^{L-1} K(a_{i+1} \mid a_i), \tag{13}$$

into an initial law μ and a *transition kernel* $K(a' \mid a)$, which for each a is a density in a'. On the grid of (3) the kernel becomes a matrix $K_{kj} = K(u_k \mid u_j)$ whose columns satisfy $\sum_k w_k K_{kj} = 1$, and pushing a distribution one step forward is a matrix–vector product:

$$p_{i+1}(a') = \int K(a' \mid a)\, p_i(a)\, \mathrm{d}a \quad \rightsquigarrow \quad p_{i+1}(u_k) = \sum_j K_{kj}\, w_j\, p_i(u_j).$$

3.2 The AR(1) chain

The simplest continuous chain is the first-order autoregression

$$a_{i+1} = \rho\, a_i + \varepsilon_i, \qquad \mathbb{E}\varepsilon_i = 0,\ \operatorname{Var}\varepsilon_i = q, \tag{14}$$

with innovations ε_i independent of the past. If the variance is to stay at v from site to site, $v = \rho^2 v + q$, so $q = 1 - \rho^2$ keeps it at 1. Iterating, $a_{i+k} = \rho^k a_i +$ (noise independent of a_i), so the correlation between sites k apart is $\rho^k = e^{-k/\xi}$ with *correlation length* $\xi = 1/\log(1/\rho)$ sites. At the project's frozen $\rho = 0.85$, $\xi \approx 6.2$ sites.

The innovation need not be Gaussian. A Laplace innovation, or a mixture of Gaussians, gives a chain with *exactly the same covariance* as the Gaussian one and different higher-order structure. That was the project's first laboratory: hold the second-order structure fixed and vary the rest.

3.3 Stationarity and reversibility

A density π is *stationary* if one step of the chain leaves it unchanged, $\pi(a') = \int K(a' \mid a)\pi(a)\,\mathrm{d}a$. On the grid, π is an eigenvector of the matrix KW (W the diagonal of weights) with eigenvalue 1. For the chains we use it is unique and the chain forgets its start geometrically fast (the power method again). A chain started from π is *stationary*: every site has law π.

The chain is *reversible* if it satisfies *detailed balance*,

$$\pi(a)K(a' \mid a) = \pi(a')K(a \mid a') : \tag{15}$$

the flow of probability from a to a' equals the flow back. A film of a reversible stationary chain looks statistically the same played backwards. Detailed balance implies stationarity (integrate both sides over a).

In our project

The index i need not be time. A snapshot of a one-dimensional field (the value of a physical quantity at each of L equally spaced points along a line) is a sequence $\phi_1, \ldots, \phi_L$. If the physics couples only neighbouring points, this sequence is a Markov chain *in space*: the value at site $i+1$ depends on the values to its left only through site i. Chapters 11 and 12 show that the stochastic Allen–Cahn field is exactly of this kind, and build its kernel K.

Check yourself.

3.1 For the AR(1) chain at $\rho = 0.85$, what is the correlation between sites ten apart?

3.2 Show that detailed balance (15) implies that π is stationary.

4 Noise, scores, and Tweedie's formula

4.1 The noising channel

All noise in this project enters through one channel. A clean value a (standardised to variance one) is replaced by

$$x = \alpha\, a + \sqrt{\Delta}\, z, \qquad \alpha = e^{-t}, \quad \Delta = 1 - e^{-2t}, \quad z \sim \mathcal{N}(0,1) \text{ independent.} \tag{16}$$

The *noise level* $t \geq 0$ runs from clean ($t = 0$, $x = a$) to pure noise ($t \to \infty$, $x \approx z$). The choice $\alpha^2 + \Delta = 1$ keeps the variance at one, so the channel is called *variance preserving*. The signal-to-noise ratio is $\alpha^2/\Delta = 1/(e^{2t} - 1)$. For a sequence, each site receives its own independent z_i. The experiments use a fixed schedule of twelve levels from $t = 0.05$ to $t = 3$. (The consolidated note writes $Y = X + \sigma Z$; dividing (16) by α shows that $\sigma = \sqrt{\Delta}/\alpha$.)

Equation (16) is not arbitrary: it is the exact solution at time t of the Ornstein–Uhlenbeck process started at a, the random motion that is pulled back towards zero while being kicked by noise. We derive this in §11.1; here we only need the formula.

4.2 The score

The density of the noised data is the clean density smoothed by the channel,

$$p_t(x) = \int p(a)\, \mathcal{N}(x; \alpha a, \Delta)\, \mathrm{d}a,$$

and its *score* is the gradient of its logarithm,

$$s_t(x) = \nabla_x \log p_t(x),$$

a vector at every point x that points towards higher probability. For a Gaussian $\mathcal{N}(0, \Sigma)$ the score is $-\Sigma^{-1}x$, a linear pull towards the centre.

Why the score? A diffusion generative model creates data by running the noising backwards: start from pure noise and undo the channel step by step. The backwards dynamics has drift $x + 2s_t(x)$ (a classical result on time-reversed diffusions), so a generator needs the score at every level t. Knowing the score is also knowing the density up to a constant.

Careful

Two different objects are called "score". The *data score* $\nabla_x \log p_t(x)$ differentiates with respect to the data; it drives denoising and generation. The *parameter score* $\partial_\theta \log p_\theta(x)$ differentiates with respect to a model parameter; it defines Fisher information (§10). Always say which variable you differentiate.

4.3 Tweedie's formula: the score is a denoiser

Proposition 4.1 (Tweedie)**.** *For the channel* (16),

$$\nabla_x \log p_t(x) = \frac{\alpha\,\mathbb{E}[a \mid x] - x}{\Delta}. \tag{17}$$

Proof. Differentiate $p_t(x)$ under the integral sign, using $\nabla_x \mathcal{N}(x; \alpha a, \Delta) = \mathcal{N}(x; \alpha a, \Delta)\,(\alpha a - x)/\Delta$:

$$\nabla_x p_t(x) = \int p(a)\mathcal{N}(x; \alpha a, \Delta)\frac{\alpha a - x}{\Delta}\,\mathrm{d}a = p_t(x)\,\frac{\alpha\,\mathbb{E}[a \mid x] - x}{\Delta},$$

because $p(a)\mathcal{N}(x; \alpha a, \Delta)/p_t(x) = p(a \mid x)$ by Bayes' rule. Divide by $p_t(x)$. □

Key idea

Estimating the score of noised data and computing the best denoiser $\mathbb{E}[a \mid x]$ are the same problem. For a sequence, $\mathbb{E}[a_i \mid x_1, \ldots, x_L]$ depends on *every* observation, and for a non-Gaussian chain $p_t(x_1, \ldots, x_L)$ has no formula: it is an L-dimensional integral. The chain structure is what makes that integral computable exactly (§6), in a cost that grows linearly with L.

4.4 How errors are measured

An estimate $\hat{s}$ of the score is scored by its *relative error* pooled over the twelve levels,

$$\Big(\sum_t \sum \|\hat{s} - s\|^2 \Big/ \sum_t \sum \|s\|^2\Big)^{1/2},$$

where the inner sums run over test sequences and sites. Two reference points recur. The *isotropic map* $\hat{s} = -x$ pretends the clean data were independent standard Gaussians; it ignores all structure and scores about 0.67. The *linear floor* is the score of the covariance twin (§2.4), the best any linear method can do; at the AR(1) laboratory with Laplace innovations it scores 0.168, at the ϕ^4 base cell 0.269.

Check yourself.

4.1 Use Tweedie's formula to compute the score when $a \sim \mathcal{N}(0, 1)$. Is the answer what you expect for the law of x?

4.2 At which noise level t is the signal-to-noise ratio α^2/Δ equal to one?

5 The Kalman filter and smoother

5.1 The model

Take the Gaussian AR(1) chain (14) with $q = 1 - \rho^2$ and observe every site through the channel:

$$a_{i+1} = \rho a_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, q); \qquad x_i = \alpha a_i + \sqrt{\Delta}\, z_i. \tag{18}$$

This is a *linear Gaussian state-space model*: a hidden state evolving linearly with Gaussian kicks, seen through linear Gaussian measurements. Every conditional law in it is Gaussian, so every computation reduces to updating a mean and a variance. Two questions are natural. *Filtering*: what do we know about a_i from the observations up to i? *Smoothing*: what do we know about a_i from all of them?

5.2 Filtering: predict, then update

Suppose that after seeing $x_1, \ldots, x_{i-1}$ we believe $a_i \sim \mathcal{N}(m_i^-, P_i^-)$ (the "prior" for site i; for $i = 1$ it is the stationary $\mathcal{N}(0, 1)$). Two steps advance the belief.

Update with x_i: this is exactly §2.3. With gain $k_i = \alpha P_i^- / (\alpha^2 P_i^- + \Delta)$,

$$m_i = m_i^- + k_i\,(x_i - \alpha m_i^-), \qquad P_i = (1 - k_i\alpha) P_i^-. \tag{19}$$

Predict site $i+1$: push the belief through the dynamics. A Gaussian times ρ plus an independent Gaussian is Gaussian, so

$$m_{i+1}^- = \rho\, m_i, \qquad P_{i+1}^- = \rho^2 P_i + q. \tag{20}$$

That is the whole Kalman filter: alternate (19) and (20). The gain k_i balances two sources of information. If the prediction is sharp (P_i^- small) the gain is small and a surprising measurement moves the belief little; if the measurement is clean (Δ small) the gain approaches $1/\alpha$ and the belief jumps to x_i/α. After a few sites the variances settle to a fixed point, because the recursion for P forgets its start.

5.3 Smoothing: the Rauch–Tung–Striebel pass

The filter at site i has not used $x_{i+1}, \dots, x_L$. A second, backward pass adds them.

Proposition 5.1 (RTS smoother)**.** *Let m_i^s, P_i^s be the mean and variance of a_i given all observations. Starting from $m_L^s = m_L$, $P_L^s = P_L$ and going backwards,*

$$m_i^s = m_i + J_i\,(m_{i+1}^s - m_{i+1}^-), \qquad P_i^s = P_i + J_i^2\,(P_{i+1}^s - P_{i+1}^-), \qquad J_i = \frac{\rho P_i}{P_{i+1}^-}. \tag{21}$$

Proof. Given $x_1, \dots, x_i$, the pair (a_i, a_{i+1}) is Gaussian with $\operatorname{Var} a_i = P_i$, $\operatorname{Var} a_{i+1} = P_{i+1}^-$ and $\operatorname{Cov}(a_i, a_{i+1}) = \rho P_i$. By (11), $\mathbb{E}[a_i \mid a_{i+1}, x_{1:i}] = m_i + J_i(a_{i+1} - m_{i+1}^-)$. By the Markov property, once a_{i+1} is known the later observations say nothing more about a_i, so this is also $\mathbb{E}[a_i \mid a_{i+1}, x_{1:L}]$. Averaging over a_{i+1} given everything gives the mean in (21). The variance follows from the law of total variance: $\operatorname{Var}(a_i \mid x_{1:L}) = (P_i - J_i^2 P_{i+1}^-) + J_i^2 P_{i+1}^s$. □

Figure 1 shows both passes on one sequence. The filter lags and is noisy, because it only looks backwards; the smoother is smoother and tighter, because each site borrows from both sides.

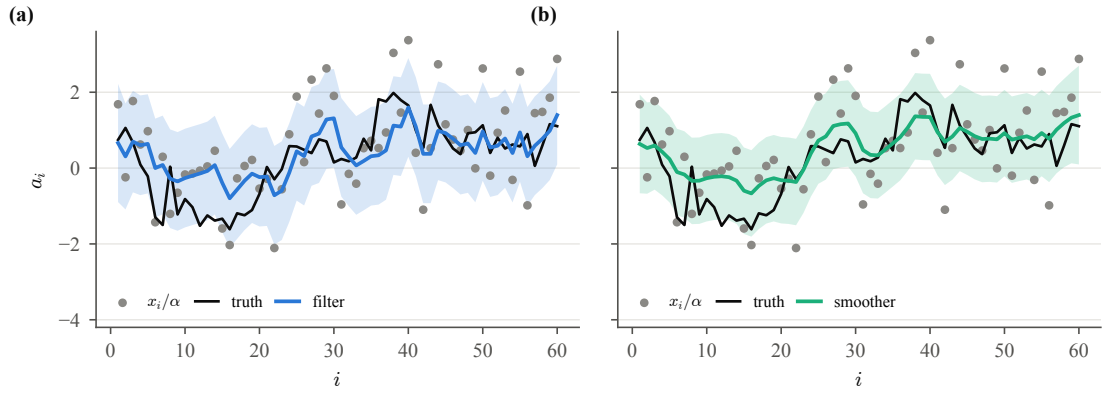


Figure 1: A Gaussian AR(1) chain ($\rho = 0.85$, black) seen through the channel at $t = 0.47$ (grey dots, rescaled by $1/\alpha$). (a) The Kalman filter mean with a two-standard-deviation band. (b) The RTS smoother: tighter, and centred on the truth rather than lagging behind it.

5.4 What the smoother computes, and what it does not

Everything in (18) is jointly Gaussian, so the conditional mean is linear in the data and equals the LMMSE estimator. Since $\operatorname{Cov}(a, x) = \alpha\Sigma_0$ and $\operatorname{Var}(x) = \alpha^2\Sigma_0 + \Delta I$ with Σ_0 the clean covariance $(\rho^{|i-j|})$,

$$m^s = \alpha\,\Sigma_0 \big(\alpha^2\Sigma_0 + \Delta I\big)^{-1} x. \tag{22}$$

The smoother computes this L-dimensional linear solve in $O(L)$ operations instead of $O(L^3)$, by exploiting the tridiagonal precision of the chain. Tweedie's formula (17) turns it into the score, and the result is the score of the Gaussian law of x itself, as it must be:

$$\frac{\alpha m^s - x}{\Delta} = \frac{\alpha^2\Sigma_0(\alpha^2\Sigma_0 + \Delta I)^{-1} - I}{\Delta}\, x = \frac{\big(\alpha^2\Sigma_0 - (\alpha^2\Sigma_0 + \Delta I)\big)(\alpha^2\Sigma_0 + \Delta I)^{-1}}{\Delta}\, x = -(\alpha^2\Sigma_0 + \Delta I)^{-1} x.$$

Now run the same recursions on a *non-Gaussian* AR(1) chain, for instance with Laplace innovations of variance q. The prediction step (20) is still correct for the mean and variance, because the first two moments of $\rho a + \varepsilon$ depend on ε only through its mean and variance. What is lost is everything else about the belief: its shape. Running Kalman on a non-Gaussian chain is running it on the chain's covariance twin, and it returns (22) again.

In our project

The paper's Proposition 1 makes this exact: *moment-projected* belief propagation (replace every message by the Gaussian with its mean and variance after each step, also called Gaussian closure or assumed-density filtering) returns exactly the LMMSE estimator of the covariance twin, whatever the innovation law. So "Gaussian closure" and "the linear floor" are the same arm. On the ϕ^4 field the projected recursion is not even well-posed (§6.5), and the closure arm is the floor by definition.

Careful

The forward pass alone is the Kalman *filter*; the full forward–backward computation is a *smoother*. The consolidated note asks that the two never be confused in the paper.

Check yourself.

5.1 Take $P^- = 1$ and the level where $\alpha^2 = \Delta = \frac{1}{2}$. Compute the gain k and the filtered variance P.

5.2 Why can the smoothed variance P_i^s never exceed the filtered variance P_i?

6 Hidden Markov models, Markov random fields, and belief propagation

6.1 Forward and backward messages

A *hidden Markov model* is any chain (13) whose states are seen only through noisy observations x_i with likelihood $\ell_i(a) = p(x_i \mid a_i = a)$; for our channel $\ell_i(a) = \mathcal{N}(x_i; \alpha a, \Delta)$. The states can be discrete (weather: sunny or rainy) or continuous (our chains). We want $p(a_i \mid x_{1:L})$ for every i without integrating over all L states at once. Define the forward and backward messages

$$\overrightarrow{m}_i(a) = p(x_1, \dots, x_i,\ a_i = a), \qquad \overleftarrow{m}_i(a) = p(x_{i+1}, \dots, x_L \mid a_i = a). \tag{23}$$

Splitting the sequence at site i and using the Markov property,

$$p(a_i = a \mid x_{1:L}) = \frac{\overrightarrow{m}_i(a)\, \overleftarrow{m}_i(a)}{p(x_{1:L})}. \tag{24}$$

Both messages obey one-step recursions. To get $\overrightarrow{m}_{i+1}$, sum over the value of a_i (the intermediate state), move one step with K, and multiply by the new likelihood:

$$\overrightarrow{m}_1(a) = \mu(a)\ell_1(a), \qquad \overrightarrow{m}_{i+1}(a') = \ell_{i+1}(a') \int K(a' \mid a)\, \overrightarrow{m}_i(a)\, \mathrm{d}a. \tag{25}$$

Symmetrically, starting from $\overleftarrow{m}_L \equiv 1$,

$$\overleftarrow{m}_i(a) = \int K(a' \mid a)\, \ell_{i+1}(a')\, \overleftarrow{m}_{i+1}(a')\, \mathrm{d}a'. \tag{26}$$

The integrals are the matrix products (4) on the grid: the cost is L multiplications of an N_g-vector by an $N_g \times N_g$ matrix, $O(LN_g^2)$, instead of the N_g^L terms of brute-force summation. In practice each message is rescaled after every step and the logarithms of the scales are accumulated, which keeps everything in range and gives the *evidence* for free: $p(x_{1:L}) = \int \overrightarrow{m}_L(a)\, \mathrm{d}a$.

Example 6.1 (three days of umbrellas)**.** A two-state chain: each day is rainy (R) or sunny (S), the weather stays the same with probability 0.8, and day one is R or S with probability $\frac{1}{2}$. We never see the weather, only whether a colleague carries an umbrella: with probability 0.9 on a rainy day and 0.2 on a sunny one. Over three days we see umbrella, umbrella, no umbrella. The recursions (25)–(26) with sums in place of integrals give:

	day 1	day 2	day 3
forward $(\overrightarrow{m}(\mathrm{R}), \overrightarrow{m}(\mathrm{S}))$	(0.450, 0.100)	(0.342, 0.034)	(0.0280, 0.0765)
backward $(\overleftarrow{m}(\mathrm{R}), \overleftarrow{m}(\mathrm{S}))$	(0.199, 0.149)	(0.240, 0.660)	(1, 1)
filtered $P(\mathrm{R} \mid \text{data so far})$	0.818	0.910	0.268
smoothed $P(\mathrm{R} \mid \text{all data})$	0.858	0.785	0.268

For instance $\overrightarrow{m}_2(\mathrm{R}) = 0.9\times(0.8\times0.450+0.2\times0.100) = 0.342$. At every day the product $\overrightarrow{m}(\mathrm{R})\overleftarrow{m}(\mathrm{R})+\overrightarrow{m}(\mathrm{S})\overleftarrow{m}(\mathrm{S})$ equals the same number, 0.1045: the evidence, the probability of the three observations. Day two shows what smoothing adds: knowing only the past, rain is 91 % likely; the dry third day lowers it to 79 %. The future revises the past, and that is the backward message's job.

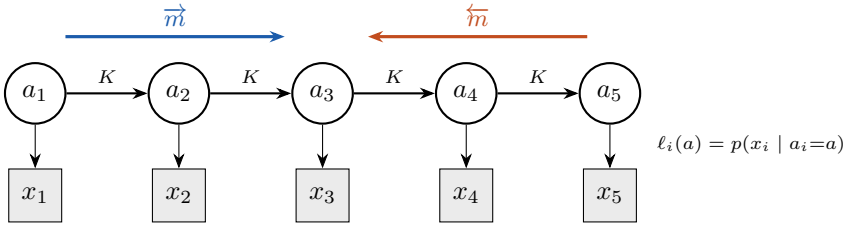


Figure 2: A hidden Markov chain. Forward messages carry the evidence from the left, backward messages the evidence from the right; at each site their product is the posterior (24).

Pair marginals. Learning (§9) needs the posterior of *neighbouring pairs*, which the same messages give:

$$\xi_i(a,a') = p(a_i = a, a_{i+1} = a' \mid x_{1:L}) = \frac{\overrightarrow{m}_i(a)\,K(a' \mid a)\,\ell_{i+1}(a')\,\overleftarrow{m}_{i+1}(a')}{p(x_{1:L})}. \tag{27}$$

Continuous states on a grid. Replace every integral by the weighted sum (3). The result is *exact* belief propagation for the discretised model, and the discretisation error is measured, not assumed: on Gaussian chains the grid reproduces the closed-form smoother (21) to double precision, and against a four-times finer grid the score changes by less than 10^{-5} relative on ordinary data. When the chain is Gaussian, every message in (25)–(26) is a Gaussian bump and the recursions reduce to the Kalman filter and smoother: grid BP carries 401 numbers where 2 would do, which is why it is a diagnostic there and a necessity elsewhere.

6.2 Markov random fields: the undirected view

The same chain can be written without arrows. A *Markov random field* on a graph assigns to every configuration an energy built from local terms and sets

$$p(a) = \frac{1}{\mathcal{Z}} \exp\big(-\mathcal{E}(a)\big), \qquad \mathcal{E}(a) = \sum_{\text{edges } (i,j)} W(a_i,a_j) + \sum_i U(a_i), \tag{28}$$

with $\mathcal{Z} = \int e^{-\mathcal{E}}$ the *partition function*. The pair energy W rewards compatible neighbours, the site energy U favours some values. The defining property is local: given its neighbours, a site is independent of everything else, because only the terms touching it change when it changes. (The Hammersley–Clifford theorem says that for positive densities this local property and the factorisation (28) are equivalent.) A *hidden* Markov random field adds noisy observations of each site; since the likelihood is a product of site terms, the posterior is again a Markov random field with U replaced by $U - \log \ell_i$. Denoising is inference in a hidden Markov random field.

On a two-dimensional grid (images) the graph has loops and exact inference is hopeless; one uses approximations such as Besag's iterated conditional modes, Gibbs sampling (Geman and Geman) or loopy belief propagation. On a line there are no loops, and everything is exact.

Transfer matrices. On a chain, write $T(a,a') = e^{-W(a,a')-U(a)/2-U(a')/2}$ (splitting each site term between its two bonds). Then the partition function is a product of matrices,

$$\mathcal{Z} = \int\!\cdots\!\int T(a_1,a_2)T(a_2,a_3)\cdots T(a_{L-1},a_L)\,(\text{boundary terms}),$$

each integral being the sum over an intermediate index in (4). Belief propagation, the forward–backward algorithm and the transfer-matrix method of statistical physics are *one computation in three languages*.

6.3 From undirected back to directed: the hidden future factor

A chain-shaped random field is a Markov chain, but its transition kernel is *not* simply $e^{-W(a,\cdot)}$ normalised. Condition on a_i: the law of a_{i+1} is proportional to $T(a_i, a_{i+1})$ times the total weight of all ways the chain can continue from a_{i+1} to the right end, which is a backward message of the unobserved chain. Arriving at a value from which many good continuations exist is more likely. For a long chain this "future factor" becomes the leading eigenvector of T (§12). Keep this in mind: it is the origin of the tilt in the tilted walk (§13).

6.4 Belief propagation in general

On any graph without loops (a tree), *sum-product* belief propagation passes messages along edges: the message from a node to a neighbour is the sum, over the node's value, of its local factor times the pair factor times all messages it received from its *other* neighbours. After one sweep inwards and one outwards every node holds its exact marginal. On a chain this is (25)–(26). Replacing sums by maximisations gives *max-product* (the Viterbi algorithm), which finds the single most probable configuration rather than the posterior mean. Drawing a_L from its posterior and then a_i from $p(a_i \mid a_{i+1}, x) \propto \overrightarrow{m}_i(a_i) K(a_{i+1} \mid a_i)$ going leftwards samples whole configurations from the posterior (forward filtering, backward sampling).

6.5 When Gaussian closure breaks

Gaussian closure replaces each message by a Gaussian with its mean and variance. For a message that says "the next site is in one of two wells, $+1$ or -1", the Gaussian with the same moments is a broad bump centred at zero, on the barrier, where the field almost never is. On the ϕ^4 chain something stronger happens: from any value far outside the bulk, the kernel jumps straight back into the bulk, so the backward message $\overleftarrow{m}_i(a)$ is nearly flat over the whole grid and its moment projection is meaningless. Measured: 38–43 % of a projected message's mass sits in the outer 5 % of the grid at the base cell and 98–100 % on the other cells (E32, E33 oracles). This is why the closure arm of the field experiments is the linear floor.

Check yourself.

6.1 Brute force would sum over every configuration of the hidden chain. For $L = 32$ sites on a grid of $N_g = 401$ points, compare the number of terms with the cost of belief propagation.

6.2 In the umbrella example, recompute $\overleftarrow{m}_2(\mathrm{R})$ from (26) and check that $\overrightarrow{m}_2(\mathrm{R})\overleftarrow{m}_2(\mathrm{R}) + \overrightarrow{m}_2(\mathrm{S})\overleftarrow{m}_2(\mathrm{S})$ is the evidence.

7 Fourier analysis: why noise erases fine detail

7.1 Waves

A complex number is $u + iv$ with $i^2 = -1$. Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ packs a cosine and a sine into one exponential of modulus one. The function $x \mapsto e^{ikx}$ is a *wave* of frequency k: it oscillates $k/2\pi$ times per unit length. Large k means fast oscillation, fine detail.

The *Fourier transform* decomposes a function into waves:

$$\hat{f}(k) = \int f(x)\, e^{-ikx}\, \mathrm{d}x, \qquad f(x) = \frac{1}{2\pi} \int \hat{f}(k)\, e^{ikx}\, \mathrm{d}k. \tag{29}$$

$|\hat{f}(k)|$ says how much of frequency k the function contains. For a density p, $\hat{p}(k) = \mathbb{E}[e^{-ika}]$ is the *characteristic function*, and its logarithm is the cumulant expansion of §2.5 with $s = -ik$. On a grid of N_g points the discrete version is computed by the fast Fourier transform (FFT) in $O(N_g \log N_g)$ operations.

7.2 Why waves: shifts, convolutions, derivatives

Waves are special because *shifting a wave only multiplies it by a number*: $e^{ik(x-c)} = e^{-ikc}e^{ikx}$. Three consequences follow, each in one line.

- **Shift.** If $g(x) = f(x - c)$ then $\hat{g}(k) = e^{-ikc}\hat{f}(k)$.
- **Convolution.** The convolution $(f * g)(x) = \int f(y)g(x-y)\,\mathrm{d}y$ is a weighted sum of shifted copies of g. Substituting $x = y + u$ in $\iint f(y)g(x-y)e^{-ikx}\,\mathrm{d}y\,\mathrm{d}x$ splits it into $\int f(y)e^{-iky}\,\mathrm{d}y \int g(u)e^{-iku}\,\mathrm{d}u$:

$$\widehat{f * g} = \hat{f}\,\hat{g}. \tag{30}$$

  Convolution, which costs $O(N_g^2)$ on a grid, becomes multiplication, which costs $O(N_g)$.

- **Derivatives.** Integrating by parts, $\widehat{f'}(k) = ik\hat{f}(k)$, so $\widehat{f''}(k) = -k^2\hat{f}(k)$. On a grid the second difference (2) acts on a wave as $e^{ik(x+h)} - 2e^{ikx} + e^{ik(x-h)} = -4\sin^2(kh/2)\,e^{ikx}$, i.e. multiplication by $-\frac{4}{h^2}\sin^2(kh/2) \approx -k^2$. Waves are the eigenfunctions of the Laplacian, with eigenvalue $-k^2$.

7.3 Noise is a low-pass filter

The density of a sum of independent quantities is the convolution of their densities. So for the channel (16), by (30) and the Gaussian transform $\widehat{\mathcal{N}(0,\sigma^2)}(k) = e^{-\sigma^2 k^2/2}$ (complete the square in the exponent),

$$\hat{p}_t(k) = \hat{p}_0(\alpha k)\; e^{-\Delta k^2/2}. \tag{31}$$

The factor $e^{-\Delta k^2/2}$ crushes high frequencies: sharp features of the clean density, such as two narrow humps separated by a dip, are the first casualties. Figure 3 shows it on the ϕ^4 marginal: its two humps are gone by $t \approx 0.3$, and at $t = 0.7$ the noised density is visibly a Gaussian.

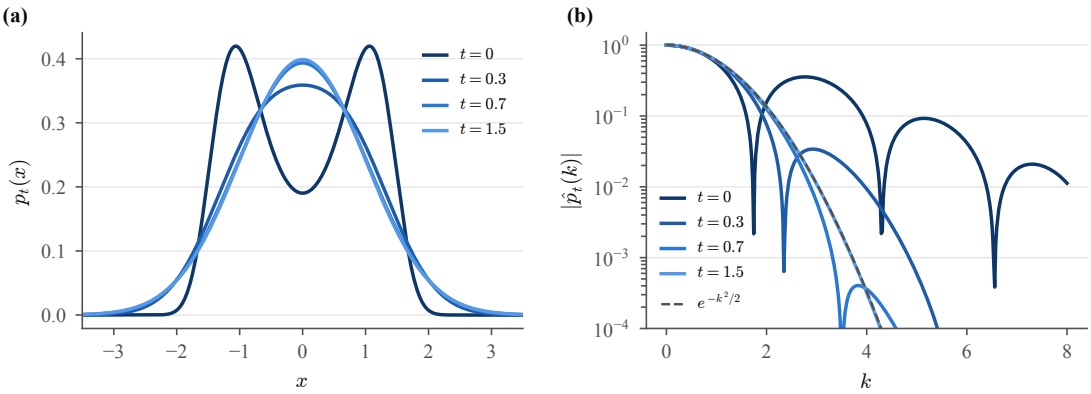


Figure 3: (a) The standardised ϕ^4 marginal at $\Theta = 0.3$, $h = 1$ ($t = 0$) and its noised versions. (b) The modulus of their Fourier transforms. The clean density has strong high-frequency content (the oscillating lobes); noise multiplies by $e^{-\Delta k^2/2}$ and leaves the Gaussian envelope $e^{-k^2/2}$ (dashed).

7.4 The heat equation

Let $u(x, s)$ evolve by the *heat equation* $\partial_s u = \frac{1}{2}\partial_x^2 u$: the value at each point moves towards the average of its neighbours, at a rate set by the second difference. In Fourier variables the right side is $-\frac{1}{2}k^2\hat{u}$, so each wave evolves on its own, $\hat{u}(k, s) = \hat{u}(k, 0)e^{-k^2 s/2}$. Comparing with (31): running the heat equation for time s is the same as convolving with $\mathcal{N}(0, s)$. The density of a randomly walking particle obeys the heat equation, and "adding Gaussian noise" and "diffusing" are the same operation. We will meet the heat equation again as the spring term of the Allen–Cahn equation.

7.5 Fourier and belief propagation

The ϕ^4 transfer kernel (§12) has the form $T(\phi, \phi') = D(\phi)\, G(\phi' - \phi)\, D(\phi')$, with G a Gaussian. A message update is then: multiply by D in real space, convolve with G (diagonal in Fourier space), multiply by D and by the likelihood (diagonal in real space). With FFTs in between, a step costs $O(N_g \log N_g)$ instead of $O(N_g^2)$. This was the meeting's "Fourier BP" idea, and the field makes it natural. We measured that it cannot pay much at our sizes: at $N_g = 401$ the dense product is a minority of the sweep time, which is dominated by memory traffic, so the end-to-end speed-up is capped near 1.3× (E24). A truncated Fourier series of a message can also go negative, which a probability cannot.

Careful

Three spectral ideas must not be confused. *Fourier modes* are a computational basis (diagonalise convolution). *Hermite modes* (next chapter) diagonalise the noising channel. *Transfer-operator eigenfunctions* (§12) describe the stationary law and the correlation length of the field.

Check yourself.

7.1 Why does adding independent noise $\sqrt{\Delta}z$ multiply the characteristic function by $e^{-\Delta k^2/2}$?

7.2 On a grid of spacing h, the fastest wave has wavelength $2h$ ($k = \pi/h$). By what factor does the second difference multiply it?

8 Hermite polynomials: why fourth-order structure fades like e^{-4t}

8.1 The right basis for the channel

The channel (16) does two things: it shrinks ($a \mapsto e^{-t}a$) and it blurs (adds Gaussian noise). Waves diagonalise blurring but not shrinking. Polynomials orthogonal with respect to the Gaussian diagonalise both. Define the inner product $\langle f, g\rangle = \mathbb{E}[f(Z)g(Z)]$ for $Z \sim \mathcal{N}(0,1)$ and orthogonalise $1, x, x^2, \dots$ one at a time (Gram–Schmidt). Using $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, $\mathbb{E}Z^6 = 15$:

$$H_0 = 1, \quad H_1 = x, \quad H_2 = x^2 - 1, \quad H_3 = x^3 - 3x, \quad H_4 = x^4 - 6x^2 + 3, \ \dots \tag{32}$$

For instance $H_2 = x^2 - \mathbb{E}Z^2$ removes the part of x^2 along 1, and $H_3 = x^3 - 3x$ removes the part along x ($\mathbb{E}[Z^3 \cdot Z] = 3$). These are the (probabilists') *Hermite polynomials* (Figure 4a). They have a compact *generating function*,

$$G(x; s) = e^{sx - s^2/2} = \sum_{n \geq 0} H_n(x) \frac{s^n}{n!}, \tag{33}$$

from which everything follows. Orthogonality: $\mathbb{E}[G(Z; s)G(Z; r)] = e^{-(s^2+r^2)/2}\mathbb{E}e^{(s+r)Z} = e^{sr}$, and matching powers of sr gives $\mathbb{E}[H_m H_n] = n!$ if $m = n$ and 0 otherwise. Differentiating (33) in x gives $H_n' = nH_{n-1}$.

8.2 The channel multiplies H_n by e^{-nt}

Proposition 8.1. *For f a function, let $(P_t f)(a) = \mathbb{E}\big[f(e^{-t}a + \sqrt{1 - e^{-2t}}\, Z)\big]$ be the average of f over the channel started at a. Then $P_t H_n = e^{-nt} H_n$.*

Proof. Apply P_t to the generating function. With $\alpha = e^{-t}$, $\Delta = 1 - \alpha^2$,

$$P_t G(\cdot; s)(a) = \mathbb{E}\, e^{s\alpha a + s\sqrt{\Delta} Z - s^2/2} = e^{s\alpha a + s^2\Delta/2 - s^2/2} = e^{(s\alpha)a - (s\alpha)^2/2} = G(a; s\alpha),$$

because $\Delta - 1 = -\alpha^2$. Expanding both sides in powers of s: $\sum_n (P_t H_n)(a) s^n/n! = \sum_n H_n(a)\alpha^n s^n/n!$. □

8.3 Densities, cumulants, and the noised law

Write a standardised density relative to the Gaussian as

$$p(a) = \varphi(a)\Big(1 + \sum_{n \geq 3} c_n \frac{H_n(a)}{n!}\Big), \qquad c_n = \mathbb{E}_p[H_n(a)], \tag{34}$$

with φ the standard normal density; the coefficients are read off by orthogonality, and $c_1 = c_2 = 0$ because the law is standardised. In particular $c_3 = \mathbb{E}a^3$ is the skewness and $c_4 = \mathbb{E}a^4 - 3$ the excess kurtosis. Now noise it. The noised quantity x has $\mathbb{E}[H_n(x)] = \mathbb{E}[(P_t H_n)(a)] = e^{-nt} c_n$ by Proposition 8.1, so

$$p_t(x) = \varphi(x)\Big(1 + \sum_{n \geq 3} c_n\, e^{-nt}\, \frac{H_n(x)}{n!}\Big). \tag{35}$$

Key idea

The channel leaves the Gaussian part alone and multiplies the n-th Hermite coefficient by e^{-nt}. Mean information decays like e^{-t}, covariance like e^{-2t}, skewness like e^{-3t}, kurtosis like e^{-4t} (Figure 4b). Fourth-order distinctions vanish under noise long before second-order ones.

8.4 What a Gaussian denoiser misses: Theorem 2 at one site

Take a symmetric law ($c_3 = 0$) and the score of (35). To first order in the small correction, $\log p_t = \log \varphi + \log(1 + R) \approx -x^2/2 + R$, so using $H_n' = nH_{n-1}$,

$$s_t(x) \approx -x + \sum_{n \geq 3} c_n\, e^{-nt} \frac{H_{n-1}(x)}{(n-1)!} = -x + \frac{\kappa_4\, e^{-4t}}{6} H_3(x) + O(e^{-6t}). \tag{36}$$

The covariance twin's score is exactly $-x$. Their mean-square distance is therefore

$$\mathbb{E}\big[(s_t - s_t^{\mathrm{G}})^2\big] \approx \frac{\kappa_4^2 e^{-8t}}{36}\, \mathbb{E}[H_3^2] = \frac{\kappa_4^2}{6}\, e^{-8t}.$$

This is the single-site version of the paper's Theorem 2, which states for a whole chain that the gap between the exact score and the Gaussian (closure) score behaves like $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, with κ_4 the tensor of fourth cumulants across sites. On the ϕ^4 field the ratio of the measured gap to this prediction rises towards one with t and reaches 0.73 to 0.94 at $t = 3$ (E32, frozen).

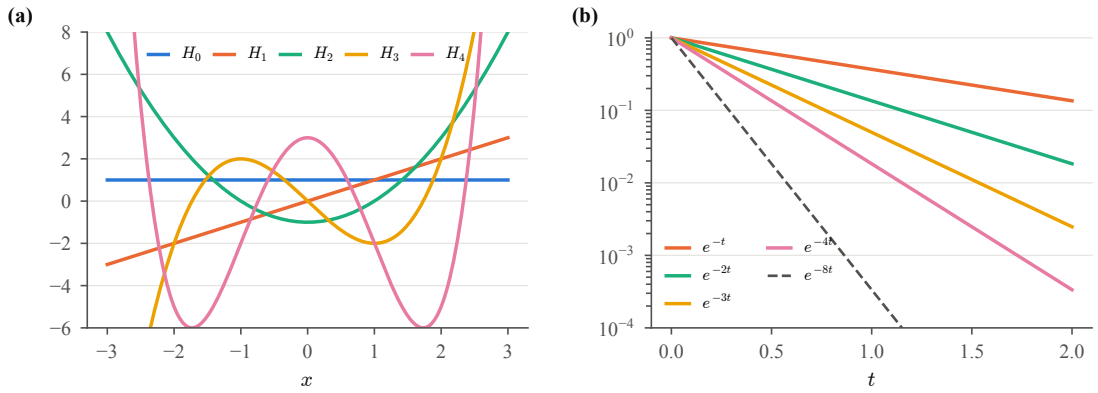


Figure 4: (a) The Hermite polynomials $H_0, \dots, H_4$. (b) The factors e^{-nt} by which the channel multiplies their coefficients, and e^{-8t}, the decay of the squared non-Gaussian part of the score.

8.5 Many sites

For a sequence, use products $H_{n_1}(x_1) \cdots H_{n_L}(x_L)$. The same proof shows that the channel multiplies such a product by $e^{-(n_1+\cdots+n_L)t}$. The covariance between sites i and j is the coefficient of $H_1(x_i)H_1(x_j)$, degree 2, so it survives like e^{-2t}; a fourth cumulant across sites has total degree 4 and fades like e^{-4t}.

Careful

Adding independent Gaussian noise σz to a does *not* change its cumulants of order 3 and higher (they add, and the Gaussian contributes zero). What shrinks is their size *relative to the variance*, which grows. Under the variance-preserving channel the two effects are combined into the single factor $\alpha^n = e^{-nt}$ of (12). Both descriptions say the same thing; mixing them up is a common source of wrong exponents.

Check yourself.

8.1 Check directly that H_3 is orthogonal to H_1 under the standard normal.

8.2 By what factor does the non-Gaussian part of the score in (36) shrink between $t = 0.5$ and $t = 1$? And its squared error?

9 Maximum likelihood, EM, and generalised EM

9.1 Likelihood and its target

A model is a family of densities p_θ indexed by parameters θ. Given independent data $y_1, \dots, y_n$, the *log-likelihood* $\mathcal{L}(\theta) = \sum_k \log p_\theta(y_k)$ scores how well θ explains them, and the *maximum likelihood estimate* (MLE) maximises it. Two examples we use. For a Gaussian, the MLE of the mean is the sample mean and of the variance the mean squared deviation. For a clean AR(1) chain, $\mathcal{L}(\rho, q) = \sum_i \log \mathcal{N}(a_{i+1}; \rho a_i, q)$, which is a least-squares problem: $\hat{\rho} = \sum_i a_i a_{i+1} / \sum_i a_i^2$ and $\hat{q}$ the mean squared residual.

What does the MLE converge to? Divide $\mathcal{L}$ by n: by the law of large numbers it approaches $\mathbb{E}_{\text{true}}[\log p_\theta(y)]$. Compare with the *Kullback–Leibler divergence*

$$\mathrm{KL}(p \,\|\, q) = \int p \log \frac{p}{q} \;\geq 0, \qquad \text{with equality only if } p = q,$$

(the inequality is Jensen's: $\mathbb{E}_p \log(q/p) \leq \log \mathbb{E}_p(q/p) = \log 1 = 0$). Since $\mathbb{E}_{\text{true}} \log p_\theta = -\mathrm{KL}(p_{\text{true}} \| p_\theta) + \text{const}$, maximising likelihood is minimising the divergence from the truth to the model.

Key idea

If the truth is in the family, the MLE finds it (consistency). If it is not, the MLE converges to the member *closest* to the truth in KL divergence, and stays there: the remaining error is an asymptotic bias that no amount of data removes. Separating "needs more data" from "needs a better family" is what several of our predictions test (E33 P1, E35 P2).

9.2 Latent variables and the evidence lower bound

Often the model describes a pair (a, x) but we observe only x: the clean chain behind the noised one, the component label behind a mixture draw. The likelihood is then the *evidence* $\log p_\theta(x) = \log \int p_\theta(x, a)\,\mathrm{d}a$, the logarithm of an integral, awkward to maximise. The way in is a lower bound. For *any* density $q(a)$, by Jensen's inequality,

$$\log p_\theta(x) = \log \int q(a) \frac{p_\theta(x, a)}{q(a)}\,\mathrm{d}a \;\geq\; \int q(a) \log \frac{p_\theta(x, a)}{q(a)}\,\mathrm{d}a =: F(q, \theta). \tag{37}$$

F is the *evidence lower bound* (ELBO). The gap is exactly a divergence: writing $p_\theta(x, a) = p_\theta(a \mid x) p_\theta(x)$ inside F gives

$$\log p_\theta(x) = F(q, \theta) + \mathrm{KL}\big(q \,\|\, p_\theta(\cdot \mid x)\big), \tag{38}$$

so the bound is tight exactly when q is the posterior.

9.3 The EM algorithm

EM climbs F by alternating over its two arguments.

- **E-step.** Set $q = p_{\theta_{\text{old}}}(a \mid x)$, the posterior under the current parameters. By (38) this closes the gap: $F(q, \theta_{\text{old}}) = \log p_{\theta_{\text{old}}}(x)$.
- **M-step.** Maximise $F(q, \theta)$ over θ with q fixed. The term $-\int q \log q$ does not involve θ, so this maximises

$$Q(\theta \mid \theta_{\text{old}}) = \mathbb{E}_{a \sim p_{\theta_{\text{old}}}(\cdot \mid x)}\big[\log p_\theta(x, a)\big], \tag{39}$$

  the expected *complete-data* log-likelihood.

Proposition 9.1 (EM never decreases the evidence)**.** $\log p_{\theta_{\text{new}}}(x) \geq \log p_{\theta_{\text{old}}}(x)$.

Proof. $\log p_{\theta_{\text{new}}}(x) \geq F(q, \theta_{\text{new}}) \geq F(q, \theta_{\text{old}}) = \log p_{\theta_{\text{old}}}(x)$: the bound, then the M-step, then the tight E-step. □

The proof also shows what is *needed*: any θ_{new} with $Q(\theta_{\text{new}} \mid \theta_{\text{old}}) \geq Q(\theta_{\text{old}} \mid \theta_{\text{old}})$ works. A decrease of the evidence is therefore always a bug or a too-generous approximate step, never a feature of EM. Our code records every decrease rather than stopping on it.

Example 9.2 (a mixture of two Gaussians)**.** Data y_k come from component $c \in \{1, 2\}$ with probability w_c, then $y_k \sim \mathcal{N}(\mu_c, \sigma_c^2)$. The label is latent. The E-step computes the *responsibilities*, the posterior probabilities of the labels,

$$r_{kc} = \frac{w_c\, \mathcal{N}(y_k; \mu_c, \sigma_c^2)}{\sum_{c'} w_{c'}\, \mathcal{N}(y_k; \mu_{c'}, \sigma_{c'}^2)};$$

the M-step sets w_c to the average responsibility and μ_c, σ_c^2 to responsibility-weighted means and variances. Figure 5 shows the evidence rising at every iteration, with a long plateau: when the components overlap, EM can crawl.

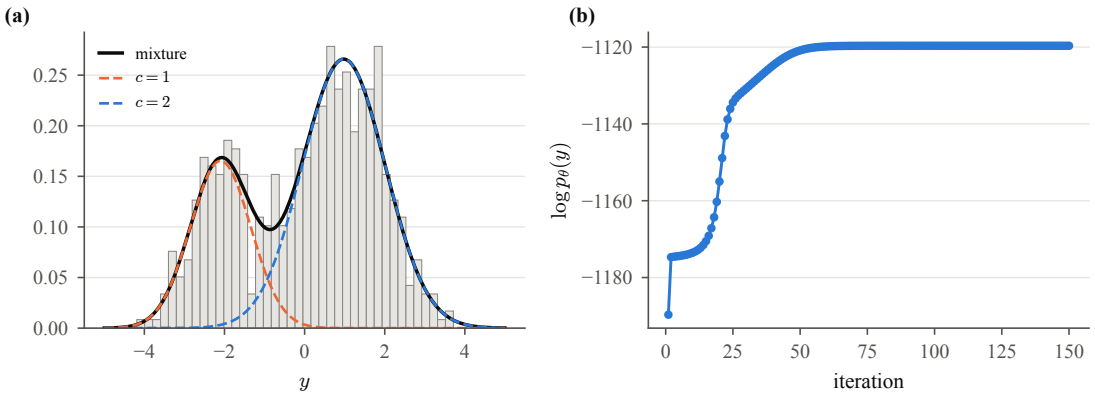


Figure 5: EM on a two-component mixture (600 draws). (a) The fitted mixture and its components. (b) The log-evidence per iteration: never decreasing, but slow while the components are still entangled.

9.4 Generalised EM: the family

Exact E- and M-steps are luxuries. The variants keep the monotone climb by relaxing one of them.

- **Generalised EM (GEM).** Any parameter with higher Q instead of the maximiser. Our kernels accept an M-step only if Q did not decrease, and otherwise return the old parameters.
- **ECM** (expectation conditional maximisation). Maximise Q over blocks of parameters in turn. The innovation-mixture kernel updates its correlation, then its components.
- **Gradient EM.** One or a few gradient steps on Q, with a line search; the neural (MDN) kernel does this with Adam steps and backtracking.
- **Variational EM.** When the exact posterior is intractable (two-dimensional fields), restrict q to a simple family and maximise F over it; the bound (37) is no longer tight, so one maximises a bound rather than the evidence.
- **Monte Carlo and stochastic EM.** Replace the expectation in (39) by an average over posterior samples; online EM updates the expected statistics as data stream in.

All are instances of *minorise–maximise*: $F(q,\cdot)$ is a function below $\log p_\theta(x)$ that touches it at θ_{old}, and any increase of the minoriser is an increase of the target.

9.5 EM on chains (Baum–Welch)

For a hidden Markov chain, $\log p_\theta(x,a) = \log \mu_\theta(a_1) + \sum_i \log K_\theta(a_{i+1} \mid a_i) + \sum_i \log \ell_i(a_i)$. Taking the posterior expectation, the only part that involves the kernel is

$$Q(\theta) = \iint \Xi(a,a') \, \log K_\theta(a' \mid a) \, \mathrm{d}a \, \mathrm{d}a' + (\text{site-one term}), \qquad \Xi = \sum_{\text{sequences}} \sum_i \xi_i, \tag{40}$$

with ξ_i the pair marginals (27). $\Xi(a,a')$ is the *expected number of transitions* from a to a', and the E-step is one forward–backward pass per sequence. The M-step is a weighted fit of the kernel to these soft counts. With clean data the posterior is a point mass and Ξ is just the histogram of observed transitions: no E-step at all.

If the kernel is an exponential family in its parameters for each source a, $\log K_\theta(a' \mid a) = \theta \cdot B(a') + c(a,a') - \log Z_\theta(a)$, then Q is a linear function of θ minus a sum of log-normalisers, hence concave, and its gradient is

$$\nabla_\theta Q = \sum_a \Big[\sum_{a'} \Xi(a,a') B(a') - N(a) \, \mathbb{E}_{K_\theta(\cdot \mid a)}[B] \Big], \qquad N(a) = \sum_{a'} \Xi(a,a') : \tag{41}$$

expected statistics under the data minus expected statistics under the model, source by source. The tilted-walk kernel of §13 is of this kind.

9.6 Fisher's identity and Boltzmann learning

Proposition 9.3 (Fisher's identity)**.** $\nabla_\theta \log p_\theta(x) = \mathbb{E}_{a \sim p_\theta(\cdot \mid x)} \big[\nabla_\theta \log p_\theta(x,a) \big]$.

Proof. $\nabla p_\theta(x) = \int \nabla p_\theta(x,a) \, \mathrm{d}a = \int p_\theta(x,a) \nabla \log p_\theta(x,a) \, \mathrm{d}a$; divide by $p_\theta(x)$. □

Three uses. The gradient of the evidence costs one E-step. At a fixed point of EM the gradient of Q vanishes, so EM converges to stationary points of the evidence. And it gives Boltzmann learning: for an energy model $p_\theta(a) = e^{-\mathcal{E}_\theta(a)} / \mathcal{Z}_\theta$ seen through a likelihood, $\nabla \log \mathcal{Z}_\theta = -\mathbb{E}_{p_\theta}[\nabla \mathcal{E}_\theta]$, so

$$\nabla_\theta \log p_\theta(x) = - \underbrace{\mathbb{E}_{a \mid x}[\nabla \mathcal{E}_\theta(a)]}_{\text{positive phase}} + \underbrace{\mathbb{E}_{a \sim p_\theta}[\nabla \mathcal{E}_\theta(a)]}_{\text{negative phase}}. \tag{42}$$

The positive phase pulls the energy down where the data's posterior sits; the negative phase pushes it up where the model already puts mass. For an undirected chain both phases are pair expectations computable by transfer matrices. For a directed kernel the negative phase becomes the per-source expectation in (41), and no chain-wide partition function appears.

9.7 How fast EM goes

Near its limit EM converges geometrically, with a rate equal to the *fraction of missing information*: the share of the complete-data information about θ that the observations do not carry (§10). When noise hides most of the clean chain, that fraction is close to one and EM needs many iterations. This is why our EM runs a ladder of checkpoints up to 1200 iterations, selects the checkpoint on held-out data, and flags runs whose selected checkpoint is the last one.

Check yourself.

9.1 Why does the M-step of the mixture produce responsibility-weighted averages?

9.2 An approximate M-step returns parameters with a *lower* Q than before. What can happen to the evidence?

10 Fisher information, Cramér–Rao, and quasi-likelihood

10.1 Fisher information

The parameter score $U_\theta(x) = \partial_\theta \log p_\theta(x)$ has mean zero at the true parameter: $\mathbb{E}U = \int \partial_\theta p_\theta = \partial_\theta \int p_\theta = 0$. Its variance is the *Fisher information*,

$$I(\theta) = \mathbb{E}\big[U_\theta(x)^2\big] = -\mathbb{E}\big[\partial_\theta^2 \log p_\theta(x)\big], \tag{43}$$

the second form following by differentiating $\mathbb{E}U = 0$ once more. It is the curvature of the expected log-likelihood at its peak: a sharp peak means the data pin the parameter down. For n independent observations it adds up to $nI(\theta)$.

Proposition 10.1 (Cramér–Rao)**.** *Any unbiased estimator $\hat\theta$ of θ from n observations has* $\operatorname{Var}(\hat\theta) \geq 1/\big(nI(\theta)\big)$.

Proof. Differentiate $\mathbb{E}_\theta \hat\theta = \theta$: $\int \hat\theta\, \partial_\theta p_\theta = 1$, i.e. $\operatorname{Cov}(\hat\theta, U) = 1$ for the total score U, whose variance is nI. By Cauchy–Schwarz, $1 = \operatorname{Cov}(\hat\theta, U)^2 \leq \operatorname{Var}(\hat\theta)\, nI(\theta)$. □

The MLE attains the bound for large n: $\sqrt{n}(\hat\theta - \theta)$ becomes Gaussian with variance $1/I(\theta)$. With several parameters I is a matrix and the bound is its inverse.

Example 10.2 (AR(1))**.** With the innovation variance q known, $\partial_\rho \log \mathcal{N}(a'; \rho a, q) = (a' - \rho a)a/q$, whose mean square under the stationary chain is $\mathbb{E}[\varepsilon^2]\mathbb{E}[a^2]/q^2 = 1/q$. So each clean transition carries information $1/(1-\rho^2)$ about ρ, and $\operatorname{Var}\hat\rho \approx (1-\rho^2)/n$ from n transitions.

10.2 Information lost to noise

When only noised data are seen, Fisher's identity says the observed score is the posterior average of the complete-data score, $U_{\text{obs}} = \mathbb{E}[U_{\text{complete}} \mid x]$. The law of total variance then splits the information:

$$I_{\text{complete}} = I_{\text{obs}} + \underbrace{\mathbb{E}\big[\operatorname{Var}(U_{\text{complete}} \mid x)\big]}_{I_{\text{missing}}}, \tag{44}$$

the *missing information principle*. Noise moves information from the first term to the second.

The channel does this selectively, and Hermite polynomials say how. A parameter that changes only the degree-n Hermite content of the law enters the noised density (35) through $c_n e^{-nt}$. Its score is then about $e^{-nt}(\partial_\theta c_n)H_n(x)/n!$, and its information about $e^{-2nt}(\partial_\theta c_n)^2/n!$. A covariance coordinate (degree 2) loses information like e^{-4t}; a shape coordinate such as the kurtosis (degree 4) like e^{-8t}. Since the number of sequences needed for a fixed accuracy is inversely proportional to the information, it grows like e^{4t} and e^{8t} respectively.

In our project

E12b measured exactly this (frozen 1 October): the sample size needed for a fixed accuracy grows with the noise level with log-slope 4.95 (interval 3.09 to 6.83) for the covariance coordinate, and for the shape coordinate the requirement is unresolved beyond $t = 0.8$ even at 8192 sequences (slope above 6.7). Both agree with the exact information curve, whose local slopes on these levels are 3.8–3.9 and 6.1–6.9; the asymptotic 4 and 8 are reached only at larger t.

10.3 Quasi-likelihood: using a wrong model on purpose

Sometimes we maximise a likelihood we know to be wrong because it is convenient. This is sound if the *estimating equation* is right. Let $g(\theta)$ be the gradient of the convenient objective. If its mean under the *true* law vanishes at the target value $\theta^\star$, the solution of $g(\hat\theta) = 0$ converges to $\theta^\star$, with variance given by the *sandwich* $A^{-1}BA^{-1}$, where $A = -\mathbb{E}[\partial g]$ and $B = \operatorname{Var} g$. When the objective is the true log-likelihood, $A = B = I$ (the information equality) and the sandwich reduces to the Cramér–Rao bound. When the model is

simply misspecified, the MLE converges to the KL projection of the truth (White's theorem) and the sandwich gives honest error bars around it.

The quasi-likelihood that matters here is the Gaussian one. For a zero-mean sequence with model covariance Σ_θ,

$$\ell(\theta) = -\tfrac{1}{2}\log\det\Sigma_\theta - \tfrac{1}{2}x^\top\Sigma_\theta^{-1}x, \qquad \mathbb{E}_{\text{true}}\,\ell(\theta) = -\tfrac{1}{2}\log\det\Sigma_\theta - \tfrac{1}{2}\operatorname{tr}\left(\Sigma_\theta^{-1}\Sigma^\star\right).$$

It depends on the data only through their covariance, and its expectation is maximised at $\Sigma_\theta = \Sigma^\star$ whatever the true law is. A Gaussian model is therefore consistent for everything the covariance determines (the correlation, the variance) and blind to everything else.

Key idea

Fitting a Gaussian-innovation kernel by EM on noised chains is a Gaussian quasi-likelihood: it converges to the covariance twin and its denoiser lands on the linear floor, never below it. To beat the floor, the kernel family must be able to represent non-Gaussian structure. That is why the project's estimator uses mixture, tilted and pair-energy kernels.

10.4 Pseudo-likelihood: avoiding the partition function

For a Markov random field the likelihood needs $\mathcal{Z}_\theta$, an integral over all configurations. Besag's *pseudo-likelihood* replaces the joint density by the product of each site's conditional law given its neighbours, $\prod_i p_\theta(a_i \mid a_{i-1}, a_{i+1})$; each factor is normalised over one site only, so no $\mathcal{Z}_\theta$ appears. It is consistent but less efficient; composite likelihoods generalise it. On a chain the exact likelihood is no harder: written with a directed kernel (13), it needs only one normaliser per source value, because the ground-state transform (§12) has absorbed the partition function. This is the special luck of one dimension that the learned pair energy exploits.

Check yourself.

10.1 For n draws from $\mathcal{N}(\mu, 1)$, what is the Fisher information about μ, and which estimator attains the Cramér–Rao bound?

10.2 Why can a Gaussian quasi-likelihood never learn the kurtosis of the innovations?

11 From a ball in a valley to the stochastic Allen–Cahn equation

11.1 Rolling downhill

Picture a ball in thick honey on a landscape of height $V(x)$. It has no inertia: its velocity is the force,

$$\frac{\mathrm{d}x}{\mathrm{d}s} = -V'(x), \tag{45}$$

where s is time. This is a *gradient flow*. Along it the height only goes down, $\frac{\mathrm{d}}{\mathrm{d}s}V(x(s)) = V'(x)\frac{\mathrm{d}x}{\mathrm{d}s} = -V'(x)^2 \le 0$, and the ball stops at a bottom. Our landscape is the *double well*

$$V(x) = \tfrac{1}{4}(x^2-1)^2, \tag{46}$$

with two valleys at $x = \pm 1$ (height 0) and a barrier at $x = 0$ of height $1/4$ (Figure 6a).

11.2 Adding random kicks

Now let the honey be warm, so that molecules kick the ball at random. A *Brownian motion* $W(s)$ is the limit of a random walk with ever smaller, ever more frequent steps: its increments over disjoint time intervals are independent and $W(s+\delta) - W(s) \sim \mathcal{N}(0, \delta)$. Its variance grows linearly with time, so its displacement grows like $\sqrt{\delta}$, much faster than δ for small δ (this is why its paths are rough). The *Langevin equation* adds kicks of strength set by a *temperature* Θ:

$$\mathrm{d}x = -V'(x)\,\mathrm{d}s + \sqrt{2\Theta}\;\mathrm{d}W, \quad \text{meaning} \quad x_{n+1} = x_n - V'(x_n)\,\delta + \sqrt{2\Theta\delta}\;z_n \tag{47}$$

for small time steps δ and independent standard normal z_n. At low temperature the ball rattles in one valley and only rarely jumps the barrier; the typical waiting time grows like $e^{\Delta V/\Theta}$ with ΔV the barrier height (the Arrhenius law). Figure 6b shows the hops at $\Theta = 0.15$.

11.3 Where the ball spends its time

Release many independent balls and follow their density $p(x, s)$. Probability is conserved, so the density in an interval changes only by what flows across its ends: $\partial_s p = -\partial_x J$, with J the *probability current*. Two things make probability flow. The drift carries density p at velocity $-V'$, a current $-V'p$. The kicks spread it from crowded to empty places, a current $-\Theta\,\partial_x p$ (Fick's law; the coefficient Θ is fixed by matching the spread of $\sqrt{2\Theta}W$ with the heat equation of §7). Hence the *Fokker–Planck equation*

$$\partial_s p = \partial_x\big(V'p\big) + \Theta\,\partial_x^2 p. \tag{48}$$

At equilibrium nothing flows, $J = 0$, i.e. $\Theta p' = -V'p$, whose solution is

$$\pi(x) = \frac{1}{\mathcal{Z}}\,e^{-V(x)/\Theta}. \tag{49}$$

This is the *Boltzmann–Gibbs* law: low energy is likely, and temperature sets how much the law cares about energy (Figure 6c). The zero-current condition is the continuous form of detailed balance (15): the Langevin dynamics is reversible.

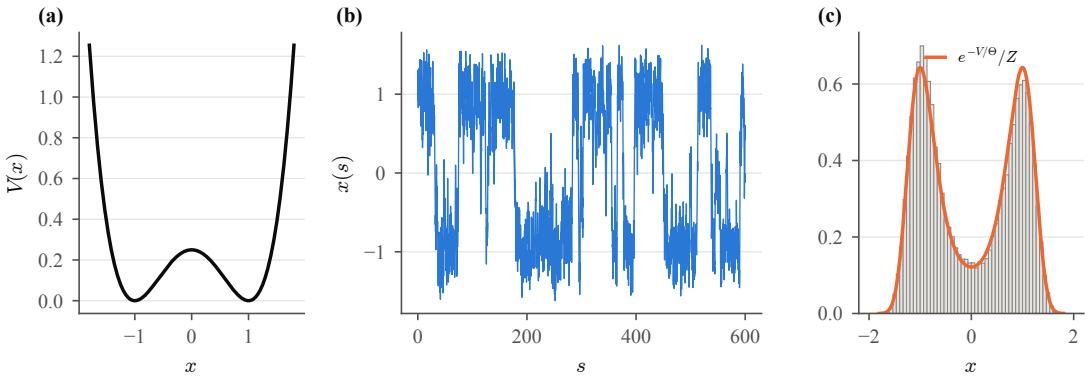


Figure 6: (a) The double well V. (b) A Langevin path at $\Theta = 0.15$: long stays in a valley, rare hops. (c) The histogram of the path against the Gibbs law $e^{-V/\Theta}/\mathcal{Z}$.

Key idea

The noising channel is a Langevin equation. With $V(x) = x^2/2$ and $\Theta = 1$, (47) is the Ornstein–Uhlenbeck process $\mathrm{d}x = -x\,\mathrm{d}s + \sqrt{2}\,\mathrm{d}W$. Multiplying by e^s gives $\mathrm{d}(e^s x) = \sqrt{2}e^s\,\mathrm{d}W$, so started at a, $x(t) = e^{-t}a + \sqrt{2}\int_0^t e^{-(t-r)}\,\mathrm{d}W(r)$, a Gaussian with variance $2\int_0^t e^{-2(t-r)}\,\mathrm{d}r = 1 - e^{-2t}$. That is exactly (16). Its Gibbs law is $e^{-x^2/2}$, the standard normal, which is why pure noise is where the channel ends.

11.4 From one ball to a string: where the Laplacian comes from

Now take many balls, one at each point $x_i = ih$ of a line with spacing h, and call the value at point i ϕ_i (think of ϕ_i as the local magnetisation, or the local composition of an alloy). Each ϕ_i sits in its own double well, and neighbours are joined by springs that dislike differences. The total energy is

$$\mathcal{E}_h(\phi) = \sum_i \Big[\frac{(\phi_{i+1} - \phi_i)^2}{2h} + h\,V(\phi_i)\Big]. \tag{50}$$

The factors of h make each piece represent a length h of string: the spring term approximates $\int \frac{1}{2}(\partial_x\phi)^2\,\mathrm{d}x$ and the well term $\int V(\phi)\,\mathrm{d}x$. The derivative of $\mathcal{E}_h$ with respect to one value is

$$\frac{\partial \mathcal{E}_h}{\partial \phi_i} = -\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h} + h\,V'(\phi_i),$$

and the second difference (2) has appeared on its own: *the springs pull each value towards the average of its neighbours*. The gradient flow of the string, with each bead's force divided by the length h it represents, is

$$\frac{\mathrm{d}\phi_i}{\mathrm{d}s} = \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} - V'(\phi_i). \tag{51}$$

As $h \to 0$ the values become a function $\phi(x, s)$ of position and time (a *field*; "scalar" only says that its value at each point is a single number), the second difference becomes the second derivative in space, and we get the *Allen–Cahn equation*

$$\partial_s \phi = \partial_x^2 \phi - V'(\phi). \tag{52}$$

It is a *partial differential equation* (PDE): an equation for a function of two variables involving its derivatives in both. Two forces compete. The spring term is the heat equation: it smooths, pulling every point towards its neighbours. The well term pushes every point towards $+1$ or -1. Allen and Cahn wrote it in 1979 for the separation of an alloy into regions of two phases.

11.5 Kinks

Where the field must change from -1 to $+1$ it forms a *kink*. A kink at rest solves (52) with $\partial_s \phi = 0$: $\phi'' = V'(\phi)$. Multiply by ϕ'; the left side is the derivative of $\frac{1}{2}\phi'^2$ and the right side the derivative of $V(\phi)$, so $\frac{1}{2}\phi'^2 = V(\phi)$ (both vanish far from the kink). Hence $\phi' = \sqrt{2V(\phi)} = (1 - \phi^2)/\sqrt{2}$, which integrates to

$$\phi_{\text{kink}}(x) = \tanh\big(x/\sqrt{2}\big), \qquad E_{\text{kink}} = \int \Big[\tfrac{1}{2}\phi'^2 + V(\phi)\Big]\,\mathrm{d}x = \int_{-1}^{1} \sqrt{2V(\phi)}\,\mathrm{d}\phi = \frac{2\sqrt{2}}{3} \approx 0.94. \tag{53}$$

The kink has width about $\sqrt{2}$ and costs a fixed energy E_{kink} (Figure 7a). A field made of long stretches near $+1$ or -1 (*domains*) separated by kinks is the typical low-temperature picture.

11.6 Adding noise everywhere: the stochastic Allen–Cahn equation

Heat the string: give every bead its own independent kicks, with strength scaled so that each piece of string of length h receives the right amount,

$$\mathrm{d}\phi_i = \Big[\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} - V'(\phi_i)\Big]\,\mathrm{d}s + \sqrt{\frac{2\Theta}{h}}\;\mathrm{d}W_i. \tag{54}$$

As $h \to 0$ this is the *stochastic* Allen–Cahn equation $\partial_s \phi = \partial_x^2 \phi - V'(\phi) + \sqrt{2\Theta}\,\eta$, with η "space-time white noise" (independent kicks at every point and every time). A PDE with random forcing is a *stochastic PDE*, an SPDE. In one space dimension this one is well defined: its solutions are continuous but rough, like a Brownian path.

Where does the string spend its time? Exactly as for one ball. For any system $\mathrm{d}X = -M\nabla\mathcal{E}\,\mathrm{d}s + \sqrt{2\Theta M}\,\mathrm{d}W$ with a positive mobility M ($M = 1/h$ here), setting every component of the probability current to zero gives

$$p(\phi_1, \dots, \phi_L) \propto e^{-\mathcal{E}_h(\phi)/\Theta} = \prod_i \exp\Big(-\frac{(\phi_{i+1} - \phi_i)^2}{2\Theta h}\Big) \prod_i \exp\Big(-\frac{h\,V(\phi_i)}{\Theta}\Big). \tag{55}$$

Key idea

The stationary law of the stochastic Allen–Cahn field on a lattice is (55): a Gaussian random walk in space (each step $\phi_{i+1} - \phi_i$ has variance Θh) reweighted at every site by $e^{-hV(\phi_i)/\Theta}$. Every factor couples at most two neighbours. So by §6 the spatial snapshot is a Markov random field on a line, hence a Markov chain in the site index, and its exact denoiser is computable by belief propagation. We never need to simulate the time dynamics (54): the stationary law is all the data are drawn from.

Two features of the lattice matter later. Each single step is *nearly Gaussian* when h is small, because the reweighting factor $e^{-hV/\Theta}$ is close to one; the non-Gaussian structure accumulates over the number of sites spanned by a correlation length. And a fixed physical length contains more sites as h shrinks, so a denoiser that looks at a fixed number of neighbours sees a shrinking piece of physics.

Check yourself.

11.1 Verify that $\tanh(x/\sqrt{2})$ solves $\phi'' = V'(\phi) = \phi^3 - \phi$.

11.2 What is the Gibbs law (49) of the Langevin equation with $V(x) = x^2/2$ at temperature Θ?

12 The transfer operator: the field as a Markov chain

12.1 The transfer kernel and its spectrum

Split each site factor of (55) evenly between the two bonds that touch it. The law becomes a product of identical symmetric factors

$$\mathsf{T}(\phi,\phi') = D(\phi)\,G(\phi'-\phi)\,D(\phi'), \qquad D(\phi)=e^{-hV(\phi)/2\Theta}, \quad G(u)=e^{-u^2/2\Theta h}. \tag{56}$$

Read T as an integral operator, $(\mathsf{T}f)(\phi)=\int \mathsf{T}(\phi,\phi')f(\phi')\,\mathrm{d}\phi'$; on a grid it is a symmetric matrix with positive entries. By §1 it has real eigenvalues $\lambda_0>\lambda_1\ge\lambda_2\ge\dots$, orthonormal eigenfunctions $\psi_0,\psi_1,\dots$, and (Perron–Frobenius) a leading eigenfunction ψ_0 that is positive everywhere. Integrating a long chain is applying T many times, so λ_0 and ψ_0 dominate.

12.2 The ground-state transform

Consider a site deep inside a long chain. Its marginal is the product of what the left half and the right half say about it. Each half is T applied many times to the constant function at the far end, which by the power method is proportional to ψ_0. So the stationary marginal is

$$\pi(\phi)=\psi_0(\phi)^2 \qquad \Big(\textstyle\int\psi_0^2=1\Big). \tag{57}$$

Given the current site, the next one is drawn in proportion to the bond factor times everything to its right, $\mathsf{T}(\phi,\phi')\psi_0(\phi')$ (the "future factor" of §6.3). Normalising with the eigen-equation $\int\mathsf{T}(\phi,\phi')\psi_0(\phi')\,\mathrm{d}\phi'=\lambda_0\psi_0(\phi)$ gives the transition kernel

$$\boxed{K(\phi'\mid\phi)=\frac{\mathsf{T}(\phi,\phi')\,\psi_0(\phi')}{\lambda_0\,\psi_0(\phi)}} \tag{58}$$

the *ground-state transform* (the name comes from quantum mechanics, where ψ_0 is the lowest-energy state). It is reversible with respect to $\pi=\psi_0^2$: $\pi(\phi)K(\phi'\mid\phi)=\psi_0(\phi)\mathsf{T}(\phi,\phi')\psi_0(\phi')/\lambda_0$ is symmetric. Intuitively, $\psi_0(\phi')$ counts the good continuations available after arriving at ϕ'.

12.3 Correlations, and why there is no phase transition

Insert the expansion $\mathsf{T}^r=\sum_n\lambda_n^r\psi_n\psi_n^\top$ into the two-point function:

$$\mathbb{E}[\phi_i\,\phi_{i+r}]=\sum_{n\ge1}\Big(\frac{\lambda_n}{\lambda_0}\Big)^r c_n^2, \qquad c_n=\int\psi_0(\phi)\,\phi\,\psi_n(\phi)\,\mathrm{d}\phi \tag{59}$$

(the $n=0$ term is the squared mean, zero by symmetry). Correlations decay geometrically, dominated by $\rho_1=\lambda_1/\lambda_0$, which plays the role of the AR(1) coefficient ρ: the correlation length is $\xi=1/\log(\lambda_0/\lambda_1)$ sites. We fix the laboratory's second-order structure by choosing h so that $\rho_1=0.85$, the same value as the AR(1) laboratory.

Because λ_0 is strictly larger than λ_1 at every temperature, ξ is finite: a one-dimensional system never commits to one phase over an infinite length. The reason is entropy. A kink costs E_{kink} but can be placed anywhere, so a long stretch always contains some, at a density that behaves like $e^{-E_{\mathrm{kink}}/\Theta}$. The domains are therefore long but finite, of length about $e^{E_{\mathrm{kink}}/\Theta}$: a sharp crossover, not a phase transition. Our frozen measurements (E32's fixed-spacing sweep) show coarse kink densities of 0.016 and 0.066 per unit length at $\Theta=0.2$ and 0.3, a ratio of 4.1, against the Arrhenius ratio $e^{E_{\mathrm{kink}}(1/0.2-1/0.3)}=4.8$.

12.4 Exact sampling and the numerics

Since the chain is stationary with known π and K, exact samples need no simulation in time: draw $\phi_1\sim\pi$, then each next site from K. In the code, T is discretised on 2401 points over $[-12,12]$, the symmetric matrix $W^{1/2}\mathsf{T}W^{1/2}$ is diagonalised once per (Θ,h), $\log\psi_0$ is interpolated by a spline, and the field is standardised to unit variance, $a=\phi/s$ with $s^2=\mathrm{Var}_\pi\,\phi$, before the noise channel is applied. The draw from K is done by rejection: propose a Gaussian step of variance Θh and accept with probability proportional to $\psi_0(\phi')D(\phi')$, a fact that the next chapter turns into a theorem.

Careful

The number of sign changes of a sampled field is *not* its number of kinks. A rough path that crosses zero once tends to recross it many times nearby, and on a finer lattice those recrossings multiply: the raw sign-change density grows like $1/\sqrt{h}$ (measured: 0.120, 0.175, 0.247 per unit length at $h=1,0.5,0.25$;

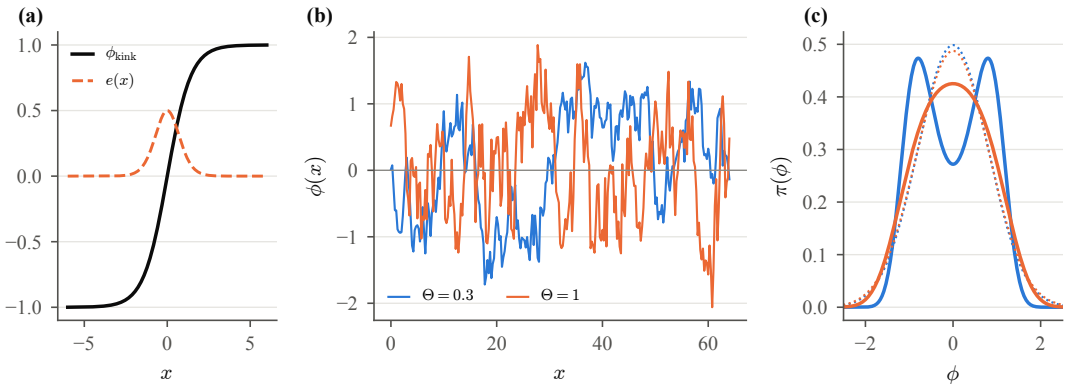


Figure 7: (a) The kink $\tanh(x/\sqrt{2})$ and its energy density, which integrates to $E_{\text{kink}} = 2\sqrt{2}/3$. (b) Exact stationary fields at $h = 0.25$ over a length of 64: at $\Theta = 0.3$ (blue) long domains near ± 1 separated by kinks; at $\Theta = 1$ (orange) a rough field that rarely settles. (c) The stationary marginal $\pi = \psi_0^2$ (solid) against the Gaussian with the same variance (dotted): bimodal at $\Theta = 0.3$, already close to Gaussian at $\Theta = 1$.

N-31). Counting only crossings that survive over a kink width ($\sqrt{2}$ in physical units) gives a statistic that agrees across lattices within 11 %, and across subsampling strides of one fine field within 8 % (E34, E36).

Check yourself.

12.1 Check that the kernel (58) integrates to one over ϕ'.

12.2 If $\lambda_1/\lambda_0 = 0.85$, after how many sites has the leading term of the correlation fallen below 0.1 of its starting value?

13 Putting it together: the tilted walk and the learned pair energy

13.1 The question, restated with everything we now have

We observe many short sequences $x_{1:L}$, each a stationary snapshot of the ϕ^4 field (§11), standardised and passed through the channel (16) at some level t. We want the exact score (17) at every level, which by Tweedie is the posterior mean, which by §6 is one forward–backward pass *if we know the kernel* K. So the whole problem is to *learn the local law* from noised data (EM, §9) and let belief propagation compose it exactly. The theory below says which families of local laws can contain the truth, what their parameters mean, and why the simpler alternatives fail.

13.2 The tilted-walk identity

Proposition 13.1 (the field's kernel is a tilted random walk)**.** *With $g = \psi_0 D$, the ground-state kernel* (58) *can be written*

$$K(\phi' \mid \phi) = \frac{\mathcal{N}(\phi'; \phi, \Theta h)\; g(\phi')}{Z_g(\phi)}, \qquad Z_g(\phi) = \int \mathcal{N}(\phi'; \phi, \Theta h)\, g(\phi')\, \mathrm{d}\phi' = \frac{\lambda_0 \psi_0(\phi)}{c\, D(\phi)}, \tag{60}$$

with $c = \sqrt{2\pi\Theta h}$ the Gaussian's normalising constant.

Proof. Insert (56) into (58): $K = D(\phi)G(\phi' - \phi)D(\phi')\psi_0(\phi')/(\lambda_0\psi_0(\phi)) = G(\phi' - \phi)g(\phi') \cdot D(\phi)/(\lambda_0\psi_0(\phi))$. The eigen-equation $\int \mathsf{T}(\phi, \phi')\psi_0(\phi')\,\mathrm{d}\phi' = \lambda_0\psi_0(\phi)$ reads $D(\phi)\int G(\phi' - \phi)g(\phi')\,\mathrm{d}\phi' = \lambda_0\psi_0(\phi)$, which is the normaliser. Replace G by $c\mathcal{N}(\cdot; \phi, \Theta h)$. □

In words: *take a Gaussian random-walk step of variance Θh, then tilt the landing point by a positive weight $g(\phi')$, and renormalise.* The weight combines the local preference $D(\phi') = e^{-hV(\phi')/2\Theta}$ with the future factor $\psi_0(\phi')$. The normaliser Z_g is the ground-state transform itself, so a model that parametrises $\log g$ and normalises each column numerically contains the truth *without ever computing ψ_0 or λ_0*. In standardised units $a = \phi/s$ the step variance is $\gamma = \Theta h/s^2$.

13.3 Where the two laboratories meet, and where they part

Proposition 13.2. *The Gaussian AR(1) kernel $\mathcal{N}(a';\rho a,q)$ with $0<\rho<1$ is a tilted walk, with $\gamma=q/\rho$ and the quadratic tilt $\log g(a')=-(1-\rho)a'^2/2q$. No other AR(1) kernel $k(a'-\rho a)$ is a tilted walk.*

Proof. For the first claim, complete the square: $\mathcal{N}(a';a,\gamma)e^{-ca'^2}$ is a Gaussian in a' with mean $a/(1+2c\gamma)$ and variance $\gamma/(1+2c\gamma)$; setting these to ρa and q gives $\gamma=q/\rho$ and $2c=(1-\rho)/q$. For the second, note that for any tilted walk the mixed derivative $\partial_a\partial_{a'}\log K=1/\gamma$ is a constant: $\log g$ depends only on a', $\log Z_g$ only on a, and the Gaussian contributes aa'/γ. For $k(a'-\rho a)$ the mixed derivative is $-\rho(\log k)''(a'-\rho a)$, constant only if $\log k$ is quadratic, i.e. k Gaussian. □

So the first laboratory (AR(1) chains with non-Gaussian innovations) and the second (gradient fields like ϕ^4) share exactly one member, the Gaussian chain. What separates them is the *drift*:

Proposition 13.3 (the drift is the gradient of the log-normaliser)**.** *For a tilted walk, $\mathbb{E}[a'-a\mid a]=\gamma\,(\log Z_g)'(a)$.*

Proof. $Z_g'(a)=\int\partial_a\mathcal{N}(a';a,\gamma)\,g(a')\,\mathrm{d}a'=\int\frac{a'-a}{\gamma}\mathcal{N}(a';a,\gamma)g(a')\,\mathrm{d}a'=Z_g(a)\,\mathbb{E}[a'-a\mid a]/\gamma$. □

For the field, $\log Z_g=\log\psi_0+hV/2\Theta+\text{const}$ (physical units), so the expected step is $\Theta h(\log\psi_0)'(\phi)+\frac{h^2}{2}V'(\phi)$: a *nonlinear* function that pushes the field back towards the wells. In the limit of small h the spatial profile obeys $\mathrm{d}\phi=\Theta(\log\psi_0)'(\phi)\,\mathrm{d}x+\sqrt{\Theta}\,\mathrm{d}B$: the drift in space is the log-derivative of the ground state, not $-V'$. An AR(1) kernel with any innovation law, including the paper's mixture of Gaussians, has a *linear* drift ρa. It is misspecified on the field through its drift, not through the shape of its noise, and no amount of data repairs that (§9).

13.4 Every tilted walk is reversible

Proposition 13.4 (reversibility and the site energy)**.** *For any $\gamma>0$ and positive g with finite Z_g, the tilted walk $K_g(a'\mid a)=\mathcal{N}(a';a,\gamma)g(a')/Z_g(a)$ is reversible with invariant density $\pi_g\propto g\,Z_g$. Equivalently, it is the Gibbs chain of the symmetric pair energy*

$$W(a,a')=\frac{(a'-a)^2}{2\gamma}+f(a)+f(a'),\qquad f=-\tfrac12\log\big(g/Z_g\big),\tag{61}$$

and conversely every pair energy of this form is a tilted walk. For the ϕ^4 field, $f(a)=\frac{h}{2\Theta}V(sa)+\text{const}$.

Proof. $\pi_g(a)K_g(a'\mid a)\propto g(a)Z_g(a)\mathcal{N}(a';a,\gamma)g(a')/Z_g(a)=g(a)g(a')\mathcal{N}(a';a,\gamma)$, which is symmetric in (a,a'): detailed balance (15), hence invariance. For the second form, take $u=\sqrt{g/Z_g}=e^{-f}$ and the symmetric operator $\mathsf{T}_g(a,a')=u(a)\mathcal{N}(a';a,\gamma)u(a')$. Then $\psi=\sqrt{gZ_g}$ satisfies $\mathsf{T}_g\psi=u\int\mathcal{N}g=uZ_g=\psi$ (using $u\psi=g$), so ψ is its Perron eigenfunction with eigenvalue 1, and its ground-state transform is $\mathsf{T}_g(a,a')\psi(a')/\psi(a)=\mathcal{N}(a';a,\gamma)g(a')/Z_g(a)=K_g$. The converse is Proposition 13.1 with $D=e^{-f}$. For the field, $g/Z_g=c\,D^2/\lambda_0$ by (60), so $f=-\log D+\text{const}=hV/2\Theta+\text{const}$. □

Key idea

A learned transition kernel of this form and a learned symmetric pair energy are *one model*. Nothing irreversible can be learned by mistake, and a fitted kernel *reads as physics*: since $\gamma=\Theta h/s^2$ and $f(a)=\frac{h}{2\Theta}V(sa)+\text{const}$,

$$\hat\Theta=\hat\gamma\,s^2/h,\qquad \hat V(\phi)=\frac{2\hat\Theta}{h}\,\hat f(\phi/s)+\text{const}.\tag{62}$$

The temperature is read from the bond variance alone, the potential (its barrier, its wells) from the site energy. Numerically, on the 401-point grid: π_g by formula and by eigensolve agree to 4×10^{-12}, detailed balance holds to 10^{-14}, and the field's own site energy equals $\frac{h}{2\Theta}V$ to 2×10^{-10}.

13.5 Three ways to learn the local law

1. **The innovation mixture** (the paper's estimator): $K(a'\mid a)=\sum_c w_c\mathcal{N}(a';\rho a+\mu_c,s_c^2)$, 25 parameters at 8 components. Exact on the AR(1) laboratory, misspecified on the field (linear drift).

2. **A tilt basis**: $\log g = \sum_m \theta_m B_m$ with Legendre polynomials (7 parameters) or cubic B-splines (25), plus $\log \gamma$. For each source value the kernel is an exponential family in θ, so the M-step is concave and its gradient is (41): posterior transition statistics minus the kernel's own conditional expectations, which is Boltzmann learning without a partition function.

3. **The site energy itself** (the physicist's parametrisation): f a quartic, five parameters with $\log \gamma$, exactly specified for ϕ^4 because $\log g = \log \psi_f - f$ is complicated but f is not. The kernel needs one eigensolve of the $N_g \times N_g$ operator T_f per evaluation, and the M-step's gradient needs the derivative of the Perron eigenvector. Differentiating $\mathsf{T}\psi_0 = \lambda_0\psi_0$ and projecting on ψ_n ($n \geq 1$) gives $\langle\psi_n, \mathrm{d}\mathsf{T}\psi_0\rangle + \lambda_n\langle\psi_n, \mathrm{d}\psi_0\rangle = \lambda_0\langle\psi_n, \mathrm{d}\psi_0\rangle$, so

$$\mathrm{d}\psi_0 = \sum_{n\geq 1} \psi_n \frac{\langle\psi_n, \mathrm{d}\mathsf{T}\,\psi_0\rangle}{\lambda_0 - \lambda_n},$$

(the component along ψ_0 vanishes because ψ_0 stays normalised). The denominators are positive by Perron–Frobenius. This objective is not concave, so a step is accepted only if Q increases.

Careful

"Exactly specified up to the basis" does not mean every fit is usable. The expected transition counts Ξ vanish beyond the data's range ($|a| \gtrsim 3.5$), so the likelihood says nothing about $\log g$ there. A global basis (Legendre) can raise the tilt in that empty region and create a pocket; the kernel's own invariant law, which the estimator uses for the first site, then sits in the pocket, and every posterior is dragged outside the data. In the previews this happened in five of eighteen Legendre fits (risks 0.18 to 1.79) and in none of fifty spline or site-energy fits. The check is the invariant law's mass beyond the data range (N-33). The site energy is immune because $g = \psi_f e^{-f}$ decays by construction.

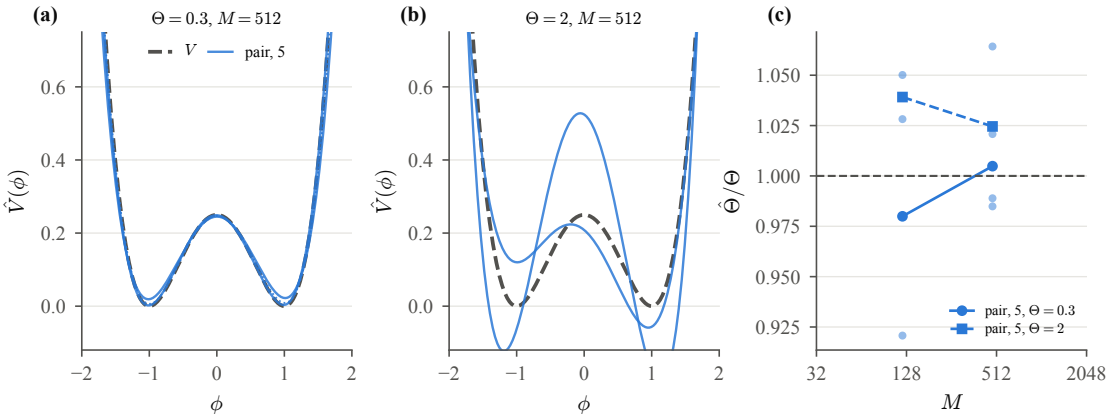


Figure 8: The physics read from kernels learned *through the noise* (laptop preview, two seeds, not yet the confirmatory run). (a) $\hat{V}$ from the five-parameter site energy at $\Theta = 0.3$, $M = 512$ sequences, against the true double well (dashed); a clean-data fit dotted. (b) The same at $\Theta = 2$, where the barrier is far below the thermal energy and the data barely constrain it. (c) $\hat{\Theta}/\Theta$ against the number of sequences.

13.6 What has been measured so far

Table 1 collects the numbers that matter for the paper, with their status. "Frozen" means the output files are checksummed in the registry; "preview" means a laptop run on two cells and at most two seeds, under the registered code but not the registered design. The confirmatory runs of E33 (sixteen seeds, six cells, five kernel arms, two network architectures) are queued on the cluster.

Read the table as one story. The linear floor (Kalman on the covariance twin, §5) leaves a quarter of the score unexplained on the base cell, because the field's structure is non-Gaussian and the Hermite argument (§8) says that structure lives in high orders. The paper's mixture kernel cannot reach it however much data it sees, because its drift is linear (§13). Kernels of the right class reach errors seven to thirteen times smaller from a few hundred noised sequences, and their parameters return the temperature and the potential that generated the data.

13.7 Open questions

- Do the confirmatory runs (sixteen seeds, six cells) confirm the preview's ordering, and does the five-parameter site energy beat the 25-parameter spline at small samples, as registered (E33 P7)?

Table 1: The main measurements on the ϕ^4 laboratory. Score errors are pooled relative errors (§4); the base cell is $\Theta = 0.3$, $h = 1$, $L = 32$.

quantity	value	source, status
Gaussian closure (linear floor), base cell	0.269	E33 oracles, 16 seeds
floor along matched correlation, $\Theta = 0.3, 0.5, 1, 2$	0.269, 0.114, 0.059, 0.041	E33 oracles
floor at fixed spacing $h = 1$, $\Theta = 0.2 \to 2$	$0.296 \to 0.184$, monotone	E32, frozen
Theorem 2 gap ratio at $t = 3$	0.73–0.94	E32, frozen
mixture kernel through noise, base, $M = 128$ and 512	0.276, 0.272 (no improvement with data)	preview
spline tilt through noise, base, $M = 128$ and 512	0.034, 0.039	preview
site energy through noise, base, $M = 128$ and 512	0.035, 0.021	preview
temperature read through noise, base, $M = 512$	$\hat{\Theta}/\Theta = 1.005 \pm 0.023$, barrier 0.234	preview
clean fits, base, $M = 2048$: site energy, spline, mixture	0.005, 0.006, 0.27	preview
samples from the learned site energy: kinks, domain length	within 5 % and 2 % of the truth	preview
coarse kinks across lattice spacings, across strides	within 11 %, within 8 %	E34, E36

- Does a window network's error grow under lattice refinement while the learned kernel's does not (E33 P5, the note's "sharp prediction")?
- Which kernel families survive subsampling, where the kernel K^s is no longer a tilted walk (E36 P3)?
- Does the augmented state repair a global latent sign (E35), completing the note's misspecification figure?
- Should the paper present the pair energy as the general symmetric $W(a, a')$ or as the gradient form (61)? The second is the whole class of discretised gradient SPDEs; the first would also cover couplings that are not of gradient type.

Check yourself.

13.1 For the Gaussian AR(1) chain at $\rho = 0.85$ with $q = 1 - \rho^2$, what are the step variance γ and the tilt $\log g$ of Proposition 13.2?

13.2 A kernel fitted at the base cell ($h = 1$, $s = 0.8207$) has $\hat{\gamma} = 0.450$. What temperature does it report?

A Notation

a_i	clean standardised value at site i	x_i	noised observation
t, $\alpha = e^{-t}$, $\Delta = 1 - e^{-2t}$	noise level and channel	$s_t(x)$	data score $\nabla_x \log p_t$
$K(a' \mid a)$	transition kernel	π	stationary density
$\overrightarrow{m}_i, \overleftarrow{m}_i$	forward, backward messages	ξ_i, Ξ	pair marginals, expected transition counts
$\ell_i(a)$	likelihood $\mathcal{N}(x_i; \alpha a, \Delta)$	N_g, u_k, w_k	grid size, points, weights
ρ	AR(1) coefficient	ξ	correlation length in sites
H_n	Hermite polynomial	c_n, κ_n	Hermite coefficients, cumulants
$Q(\theta \mid \theta')$	EM objective	$F(q, \theta)$	evidence lower bound
U_θ, $I(\theta)$	parameter score, Fisher information	KL	Kullback–Leibler divergence
ϕ_i, h, Θ	field value, spacing, temperature	V	double-well potential
$\mathcal{E}_h$	lattice energy	T, λ_n, ψ_n	transfer operator and its eigenpairs
g, Z_g, γ	tilt, normaliser, step variance	f, W	site energy, pair energy

B Where each idea lives in the code

idea	file
grid BP, evidence, posterior means	`src/markov_score/bp_grid.py`
E-step (pair marginals), EM loop, Fisher's	`src/markov_score/em.py`
identity checks	`tests/test_fisher_bruteforce.py`
Gaussian closure = LMMSE (Prop. 1), the	`tests/test_gaussian_bp_equivalence.py`
Kalman check	`bp_grid.grid_bp_projected_batch`
mixture and neural kernels	`src/markov_score/kernels.py`
ϕ^4 transfer operator, ψ_0, exact sampling	`src/markov_score/field.py`
tilted walk, closed-form π_g and site energy	`src/markov_score/tilted_kernel.py`
pair energy with quartic site energy, readings of Θ and V	`src/markov_score/pair_energy_kernel.py`
global sign latent and its repair by state augmentation	`src/markov_score/augmented.py`
the experiments of this primer	`experiments/e35_state_augmentation.py`
derivations in prose	`experiments/e32`–`e36`, registry in `experiments/registry/`
	`docs/FIELD_MODEL.md`
	`reports/field-laboratory-2026-10-01.pdf`

C Answers to the check-yourself questions

1.1 $(x+h)^3 - 2x^3 + (x-h)^3 = 6xh^2$, so the ratio is $6x = f''(x)$. The error in (2) comes from the fourth derivative, which is zero for a cubic.

1.2 Eigenvalue 3 with eigenvector $(1,1)/\sqrt{2}$, eigenvalue 1 with $(1,-1)/\sqrt{2}$. Since $3^N \gg 1^N$, $A^N v$ turns towards $(1,1)$, the positive Perron vector, unless v is exactly perpendicular to it.

2.1 Precisions add: $1/P' = 1 + 1 = 2$, so the posterior is $\mathcal{N}(x/2,\, 1/2)$. The guess splits the difference between the prior mean 0 and the observation.

2.2 $\kappa_4(x) = e^{-4} \times (-1.35) \approx -0.025$: at $t = 1$ the noised marginal is almost Gaussian.

3.1 $0.85^{10} \approx 0.197$.

3.2 Integrate both sides of (15) over a: $\int \pi(a) K(a' \mid a)\, \mathrm{d}a = \pi(a') \int K(a \mid a')\, \mathrm{d}a = \pi(a')$.

4.1 (a, x) is jointly Gaussian with $\mathrm{Cov}(a,x) = \alpha$ and $\operatorname{Var} x = 1$, so $\mathbb{E}[a \mid x] = \alpha x$ and the score is $(\alpha^2 x - x)/\Delta = -x$. Indeed $x \sim \mathcal{N}(0,1)$, whose score is $-x$.

4.2 $e^{2t} - 1 = 1$, so $t = \frac{1}{2}\log 2 \approx 0.35$.

5.1 $k = \alpha/(\alpha^2 + \Delta) = 1/\sqrt{2} \approx 0.71$ and $P = (1 - k\alpha) = \frac{1}{2}$: one measurement at this level halves the variance.

5.2 For Gaussians the posterior variance does not depend on the observed values, and conditioning on more variables cannot increase the expected posterior variance (law of total variance). So it can only shrink.

6.1 $401^{32} \approx 10^{83}$ terms against about $32 \times 401^2 \approx 5 \times 10^6$ multiplications.

6.2 $\overleftarrow{m}_2(\mathrm{R}) = 0.8 \times 0.1 \times 1 + 0.2 \times 0.8 \times 1 = 0.24$ (no umbrella on day three has probability 0.1 if rainy, 0.8 if sunny). Then $0.342 \times 0.24 + 0.034 \times 0.66 = 0.1045$.

7.1 By independence $\mathbb{E}e^{-ik(a+\sqrt{\Delta}z)} = \mathbb{E}e^{-ika}\, \mathbb{E}e^{-ik\sqrt{\Delta}z}$, and the transform of $\mathcal{N}(0,\Delta)$ is $e^{-\Delta k^2/2}$.

7.2 $-\frac{4}{h^2}\sin^2(\pi/2) = -4/h^2$, the largest damping the grid can express.

8.1 $\mathbb{E}[(Z^3 - 3Z)Z] = \mathbb{E}Z^4 - 3\mathbb{E}Z^2 = 3 - 3 = 0$.

8.2 The part proportional to e^{-4t} shrinks by $e^{-2} \approx 0.14$; its squared error by $e^{-4} \approx 0.018$.

9.1 Q is $\sum_k \sum_c r_{kc}[\log w_c + \log \mathcal{N}(y_k; \mu_c, \sigma_c^2)]$, a responsibility-weighted Gaussian log-likelihood for each component; maximising a weighted Gaussian likelihood gives weighted means and variances, and maximising $\sum r_{kc} \log w_c$ under $\sum_c w_c = 1$ gives average responsibilities.

9.2 The chain of inequalities in the monotonicity proof breaks at the middle step, so the evidence can decrease. Our kernels refuse such steps.

10.1 $U = \sum_k (y_k - \mu)$ has variance n, so $I = 1$ per draw and the bound is $1/n$; the sample mean has exactly this variance.

10.2 Its expected value depends on the true law only through the covariance; two laws with the same covariance and different kurtosis give the same objective, so nothing in it can distinguish them.

11.1 $\phi' = (1-\phi^2)/\sqrt{2}$, so $\phi'' = -\sqrt{2}\,\phi\,\phi' = -\phi(1-\phi^2) = \phi^3 - \phi$.

11.2 $\pi \propto e^{-x^2/2\Theta}$: the Gaussian $\mathcal{N}(0, \Theta)$.

12.1 $\int \mathsf{T}(\phi, \phi')\psi_0(\phi')\,\mathrm{d}\phi' = \lambda_0\psi_0(\phi)$ by the eigen-equation, which cancels the denominator.

12.2 $0.85^r < 0.1$ for $r > \log 0.1/\log 0.85 \approx 14.2$, so after 15 sites.

13.1 $\gamma = q/\rho = 0.2775/0.85 \approx 0.326$ and $\log g(a') = -(1-\rho)a'^2/2q \approx -0.270\,a'^2$.

13.2 $\hat{\Theta} = \hat{\gamma}s^2/h = 0.450 \times 0.6736 \approx 0.303$, against the true 0.3.

D Where to read more

- Probability, Gaussians, EM, graphical models: C. M. Bishop, *Pattern Recognition and Machine Learning* (2006), chapters 2, 8, 9 and 13.
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- SPDEs, for later: G. Da Prato and J. Zabczyk, *Stochastic Equations in Infinite Dimensions*.